

Post-UTME

Past Questions & Solutions

Customized
For your School of Choice

Prepare & Get an Insight on
What to expect from your
choice school's Post-UTME Exam

With 1
year SMS
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**2013/2014
Edition**

Bestselling Guide



UNIBEN POST-UTME SCREENING Past Questions & Solutions + 1 Year SMS Alerts

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PREFACE

Until the national consensus to adopt post-test by various tertiary institutions in the country, the Joint Admission and Matriculation Board-JAMB had remained the only body charged with the responsibility of testing, screening, preparing and presenting qualified secondary school leavers for admission into the over 76 universities (government and private inclusive). The introduction of the POST-UTME is not unconnected with the malpractices associated in the conduct of the jamb exams nationwide in the past five years. Thus in 2010, the name degenerated from POST-UTME to POST-UTME thereby incorporating and involving both Polytechnics and Monotechnics nationwide.

The advent of this "menace" as often seen by prospective Nigerian students was as old as 2005, today no school accepts candidates for admission without duly testing the person's ability through the post Jamb test. This therefore possess yet another hurdle and barricade to young school leavers; bridging them from getting admitted each year.

In the course of, and in preparation for this examination/test, the prospective student faces a lot of worries like; how many questions am I expected to write? What schemes and subjects do I need to meet up with? When will the application start? How am I going to be judged or graded? What are my chances of getting admitted? And of course where can I get a prototype, copy and replica of 'my choice of school' past questions and possible solutions to the questions? All these and more had remained rhetorical until the emergence of MYSCHOOL ADMISSION SUPPORT.

In our quest to remain, maintain and sustain our current status as the best and ever first corporate organization that manage the entirety of Nigerian Undergraduates, we had in the past one year embarked on massive compilation, revision, analysis and research on previous post UTME examinations conducted, the feasibility studies and chances of a candidate passing or and of course failing the test. From our research findings, we have found without further verifiable hypothesis that the POST-UTME past questions remain a secret to passing through these tests in success.

We therefore present and recommend these past questions for anyone/everyone preparing for this examination. Our idea in this sense does not focus on duplicating the efforts made by several other Nigerians, Lecturers, teachers and tutors in this direction. We only aimed at correcting some of their shortfalls, and most importantly; making these study, revision and preparatory materials readily available to anyone in any part of the country.

The questions contained herein have been drawn from the candidate's school of interest/choice in the relevant courses to facilitate easy and more result orientated performance. In this respect, actual and real examination questions were extracted from your school of choice and are therefore being presented to you, read well to succeed.

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UNIVERSITY OF BENIN, BENINCITY, NIGERIA

Post Universities Matriculation Examination 2005

Time allowed 1 hour

Read the passage below carefully and answer the questions that follow.

Each question carries 2 marks.

Comprehension Passage I

Farming is the most important aspect of agriculture that has attracted attention within the last few years. Agriculture has several other aspects like Fishery, Livestock poultry, All these are also important in that they have to do with the production of items, which human beings consume for survival.

In many parts of the world today, farming has been regarded as the main stay Of the economy, Crops such as cocoa, rubber and cotton have been produced in such commercial quantities that are sold to other countries. In these other countries, there is to spend a lot of money on agriculture, particularly farming. Most fanners use out modern tools. A lot of them have to store crops, most of which are always destroyed by insects and pests before harvest time. All these have adverse effect on their productivity.

The government can do a lot to help farmers. Farmer's co-operative societies can be encouraged and loans can be made available to farmers through government Institutions like banks and finance corporations. Farmers can be taught how to build good and strong structures for their produce. All these can help to improve the conditions of farmers in these countries.

1. The most important aspect of agriculture mentioned in the passage is (a) poultry (b) fishery (c) livestock (cl) farming.
 2. Farming in many countries today is (a) an alternative to poultry (b) of great assistance to the economy (c) for those who do 'not have jobs (d) for the illiterates.
 3. Some countries produce more and better crops than others because the fanners in the former (a) are more educated (b) have greater manpower (c) have more Modern equipment (d) have fertile land.
 4. In order to improve the state of farming, the Government should (a) give all farmers enough money to work with (b) sell enough fertilizers to all Farmers (C) find ways of financing and modernizing the farming system (d) help farmers with the storage of their crops.
 5. A lot of crops harvested are wasted because farmers (a) allow insect and pest to destroy their crops (b) do not have enough money to invest in harvesters (Q) do not have good storage facilities (d) harvest too much in a time.
- + Note: For more comprehension tips, see solution to 2006 comprehension passages.

COMPREHENSIVE PASSAGE II

Development in electronic science have transformed the art of record keeping in modern age. Traditionally, records of events were kept only in 4 people's minds. It depended very much on the retentive power of the human memory. This was extremely dangerous as people either forget events wholly or in part, or deliberately falsified details to suit their various interests. Interminable arguments were thus the order of the day. Even writing, which replaced mental recording was not entirely free from these shortcomings as untruths could be written as true records either willingly or inadvertently. With the advent of the electronic memory, however, these dangers have been largely overcome. Recordings in audio and videocassettes now show not only what happened, but also who did or said what including how and when.

6. The author believes that electronic recording is
 - (a) superior only to mental recording
 - (b) inferior to both mental recording and writing

- (c) superior to both mental recording and writing
(d) inferior to only writing.
7. The writer believes that the art keeping has
(a) improved over the years
(b) endangered the art of writing
(c) enhanced human memory
(d) overcome all the problems
8. How many stages of development did the writer mentioned while discussing the art of record keeping?
(a) Two (b) Three (c) Four (d) Five
9. According to the author, human memory is unreliable because people
(a) die and we forget what they said (b) forget events or tell lies
(c) do not always know when event happened (d) do not always know who did what and when
10. From the passage, we gather that writing is almost
(a) as unreliable as human memory
(b) as unreliable as electronic memory
(c) more reliable than the electronic memory
(d) not to be compared to any other recording system.

In each of the questions 11 to 13, identify the option that has a stress pattern different from the rest

11. (a) Factory (b) Distinctive (c) Regular (d) National.
(From the above, the only word that has a different stress pattern is "distinctive" the other three words are stressed in the first syllable. Only distinctive is stressed in the second syllable).
12. (a) Interpret (b) Period (c) Destructive (d) Forever
(The word "Period" is different from the rest in the sense that it is stressed in the first syllable, while the rest are stressed in the second syllable).
13. (a) Useless (b) Enact (c) Unless (d) Combine.
(The word "Unless" is stressed in the first syllable while the others are stressed in the second syllable).

In each of questions 14 to 26 choose the option nearest in meaning to the word(s) or phrase(s) underlined.

14. The singer's popularity is beginning to decline.
(a) slope (b) disappear (c) wane (d) darken.
("Wane" means "to diminish", "fade", "decrease").
15. Ade is at the verge of nervous breakdown, so he must be careful.
(a) brink (b) margin (c) avenue (d) fringe.
("Verge" means "very close to an extreme state or condition").
16. An open car gives no protection against the elements.
(a) weather (b) atoms (c) dust (d) atmosphere.
(Connotatively, elements refer to factors of the environment which is the weather. But denotatively, it means the building block of atoms).
17. He has never been a good mediator in family disputes because, in most cases, his views are jaundiced.
(a) coloured (b) prejudiced (c) suspicious (d) unscrupulous.
("Jaundice" means "having unpleasant notion about something due to past experience").

18. In some parts of our society, people are ostracized purely on the basis of their parentage.
(a) hated (b) disrespected (c) locked up (d) shut off from others.
("To be ostracized" means "not to be allowed to be part of a group" i.e. "shut off from others").
19. My friend will hate his Uncle forever because he left him in the lurch at his hour of need.
(a) abandoned him (b) punished him (c) disrespected him (d) despised him.
("To be left in the lurch" means "to be abandoned or deserted when one is in dire need of help"). It is an idiom.
20. All the efforts made to settle the quarrel were futile
(a) successful (b) fruitless (c) wicked (d) reasonable.
("Futile" means "unsuccessful", "fruitless", "unproductive", "pointless").
21. Kelechi hardly ever falls sick
(a) most often (b) very seldom (c) frequently (d) sometime.
("Hardly ever" means "almost never". Its synonym in the above is "very seldom"). It is a phrase.
22. It is sheer fantasy to believe that there could never be a miscalculation by any nation that has atomic weapon.
(a) dream (b) hope (c) nonsense (d) joke
("Fantasy" means "an imagined pleasant situation unlikely to be a reality", i.e. "a product of imagined situation". Its synonym is "dream").
23. The ingenious scheme of the robbers broke down because of the efficiency of the police officer.
(a) suspicious (b) ambitious (c) deceitful (d) clever.
("Ingenious" means "extremely clever").
24. Our town is so small that we are always meeting our school master accidentally.
(a) falling into (b) finding (c) running into (d) stepping on.
("Meeting" as used in the sentence means "to come in contact with something or somebody". "Running into" is the most correct).
25. One has to score not less than 200 marks in the UME together with meeting other conditions to be qualified to write the Uniben post UME screening test.
(a) to be exempted from (b) to become a candidate for
(c) to be considered for (d) to seek permission for.
("To be qualified" means "to be eligible to do something without permission").
26. He went through fire before he qualified as a doctor.
(a) a fire accident (b) a hilarious experience (c) some purification (d) a tough time.
("Fire" as used means "harsh conditions", "unpleasant situations", "tough time").

In questions 27 to 40, choose the options that are *opposite in meaning* to the words underlined.

27. The chairman's verdict redressed the injustice meted out by the secretary.
(a) corrected (b) restored (c) aggravated (d) addressed.
("To redress" means "to bring back to normal", "restore", "correct", etc. Its antonym is "aggravate" which means, "to worsen a situation").
28. Uche could not travel last December because of the severity of the harmattan.
(a) cold (b) mildness (c) warmth (d) moderation.
("Severity" as used in the sentence means "extreme". Its antonym is "mildness").
29. Travelling in the night can be ordeal
(a) a great exercise
(b) a great fear

- (a) a great endurance
(d) a great pleasure.
("Ordeal" means "very unpleasant experience").
30. Bola is very provocative in her style of dressing.
(a) modest (b) happy
(c) angry (d) beautiful.
("Provocative" means "intended to make somebody very angry or sexually excited". Its antonym is "modest" which means "not intended to attract attention especially in a sexual way").
31. Joke thought that her father was very callous.
(a) indifferent (b) compassionate
(c) wicked (d) careless.
(The adjective "callous" means "not caring that other people are suffering". Its antonym is "compassionate" which means sympathy).
32. The lions of Cameroon is really an indomitable team
(a) a powerful (b) a difficult (c) a weak (d) an unexciting.
("indomitable" means "powerful", "unbeatable", "strong" etc).
33. To almost everyone in the little village, Ada's behaviour is the most odious.
(a) repulsive (b) compassionate (c) interesting (d) charming.
(An "odious" behaviour means "an unpleasant or a repulsive one". Its antonym is "interesting" which means "the quality that something has when it attracts one's attention").
34. Adamu was so vociferous during the meeting of the congregation that he succeeded in incurring the wrath of the chairman.
(a) an ideal (b) admiration (c) displeasure (d) sympathy.
("Wrath" means "extreme anger". Its antonym is "admiration" which means, "having liking for someone").
35. It was an inopportune moment for you to make that suggestion.
(a) an ideal (b) a wrong (c) an exact (d) a sad.
(An "inopportune moment" means "a wrong", "not appropriate" or "convenient moment").
36. He devoted too much time to the peripheral aspects.
(a) unimportant (b) superficial (c) minor (d) main.
("Peripheral" means "less important part of something").
37. The climate of Nigeria is an enervating one.
(a) a weakening (b) an energy sapping (c) an invigorating (d) sluggish.
("Enervating" as an adjective means "not full of vigor or strength", "weak or tired". Its antonym is "invigorating" which means "strong and full of vigor").
38. The girl's idiosyncrasy was a passion for bread and butter.
(a) stupid outburst (b) general tendency (c) singular characteristics (d) occupational calling
("Idiosyncrasy means a behaviour", "peculiarity", "foible", "habit").
39. Once he has decided that something needs doing, my brother is very unswerving of purpose.
(a) erratic (b) tenacious (c) stubborn (d) simple.
40. Those were the days of fallacy that some languages are primitive while others are not.
(a) argument (b) exercise (c) misconception (d) truth.

ANSWERS

1. D 2. B 3. C 4. C 5. C
6. C 7. A 8. B 9. B 10. A
11. B 12. B 13. C 14. C 15. A
16. A 17. B 18. D 19. A 20. B
21. B 22. A 23. D 24. C 25. B
26. D 27. C 28. B 29. D 30. A
31. B 32. C 33. C 34. B 35. A
36. D 37. C 38. B 39. D 40. D

SECTION B: GENERAL PAPER CHEMISTRY

PHYSICS

BIOLOGY

TIME ALLOWED: 1 HOUR

- Which of the following is not true about semi conductors?
(a) Moving holes are equivalent to moving positive charges
(b) There are two kinds of charge carriers; free electron and a hole
(c) The escape of a valence electron from an atom produces electron hole pair of charge carrier
(d) Increase in temperature increases its electrical resistance.
(In a semiconductor, the electrical resistance decreases with increase in temperature).
- The minimum energy necessary to remove an electron from a given atom at infinity is called...
(a) excitation energy (b) ground state energy (c) ionization energy (d) binding energy.
(The minimum amount of energy required to remove an electron from an atom is called ionization energy).
- Find the De Broglie wavelength of a 0.01kg pellet having a velocity of 10m/s [$n = 6.63 \times 10^{-34}$ Js].
(a) 6.63×10^{-31} m (b) 6.63×10^{-32} m (c) 6.63×10^{-33} m (d) 6.63×10^{-35} m
SOL: De Broglie formula for wavelength $\lambda = h/mv$ where h is Planck's constant, m is mass and v is velocity.
 $\lambda = 6.63 \times 10^{-34} / 0.01 \times 10 = 6.63 \times 10^{-33}$ m
- A set-up transformer is designed to operate from a 25V supply. If the transformer is 80% efficient, determine the current in the primary coil when the output terminals are not connected to 240V 100w lamp.
(a) 5.0 A (b) 4.0 A (c) 2.5A (d) 2.0A.
SOL: Efficiency = Output/input x 100/1. Input power = 1pEp i.e. 80 = 100/251 px100/1, 1p= 100x 100/25x80=5A.
- An object of mass 0.2kg and density 600kgm⁻³ is suspended with a string so that 1/10 of it is immersed in paraffin of density 900kgm⁻³. Find the tension in the string.
(a) 0.2N (b) 2.0N (c) 1.0N (d) 0.1 N.
SOL: $D = M/V$. When $D = 600$, $\therefore 600 = 0.2/V$, $V = 0.2/600 = 3.3 \times 10^{-4}$
Thus 1/10 of $V = 1/10 \times 3.3 \times 10^{-4} = 3.3 \times 10^{-5}$, there when $D = 900\text{kgm}^{-3}$
 $D = m/V$ where $V = 3.3 \times 10^{-5}$, $m = 900 \times 3.3 \times 10^{-5}$, $m = 0.03$, $F = mg$, $F = 0.03 \times 10$,
 $F = 0.3$
- A rocket burns fuel at the rate of 20kgs⁻¹ and ejects it with a velocity of $5.0 \times 10^3 \text{ms}^{-1}$. Calculate the thrust exerted by gas on the rocket.
(a) $1.0 \times 10^5 \text{ms}^{-1}$ (b) $2.0 \times 10^5 \text{ms}^{-1}$
(c) $3.0 \times 10^5 \text{ms}^{-1}$ (d) $5.0 \times 10^5 \text{ms}^{-1}$.

- SOL: The thrust exerted by the gas on the rocket is the mass x velocity. i.e. $5 \times 10^5 \times 20\text{kg}^{-1} = 1.0 \times 10^5\text{ms}^{-1}$
7. Which of the following pairs consist fundamental quantities only?
- (a) Velocity and gravitational potential (b) Acceleration and field strength
(c) Momentum and work done (d) Moment and mass.
- (All are derived unit except mass; N.B
Mass, Length, Time, Temperature,
Viscosity, Amount of substance etc. are all fundamental quantities).
8. One of the limitations of Thomson's model of the atom in that it does not explain.
- (a) small angle (b) stability of the atom
(c) ionization process (d) the variation of the effective atomic radius.
- (Thomson's concluded in what is called plum pudding mode) of the atom which deals with large angle scattering and not small angle scattering).
9. A wire carrying a current of 1 GA and 2.5min length is placed in a field of flux density 0.14T. What is the force of the wire if it is placed at 60° to the field?
- (a) 30.3N (b) 20.5N (c) 15.3N (d) 10.5N
- SOL: $F = BIL\sin\theta$. Therefore, $F = 0.14 \times 2.5 \times 10\sin 60$, $F=3.03\text{N}$.
10. In the transformer, the magnetism of the core is repeatedly reversed by the magnetic field resulting in energy dissipation as heat. This loss is called.
- (a) eddy current (b) hysteresis loss
(c) flux linkage (d) joule heating loss.
- (Eddy current is minimized by lamination, making a core of soft magnetic material to reduce energy needed to facilitate reversal (Hysteresis).
11. An atom is said to be excited if an electron of the atom is ...
- (a) in the ground state (b) at infinity
(c) promoted to an energy level higher than its original level (d) having an energy value of 0.0ev.
- (An atom is excited when it is not at its original level but at a higher one).
12. What happens to the proton number Z and the nucleus number A of nuclide which emits α -radiation?
- (a) Z increases by 1 while A does not change (b) Z increases by 1 While A not change
(c) Z and A increases not decreases (d) Z increases by 1 while A decreases by 1.
- (When gamma rays are emitted the proton and nucleus number are not affected).
13. A note that is an OCTAVE higher than a given note of frequency 256Hz would have a frequency of ...
- (a) 204Hz (b) 102Hz (c) 128Hz (d) 512Hz.
- SOL: The octave of a given note $F_0=2F$, therefore $F_0 = 256$, $F_0 = 512\text{Hz}$
14. A sound wave of frequency 130Hz and wavelength 2.0m was produced at a distance, from a target and an echo was heard at the source 0.5 sec later. Calculate the values of d.
- (a) 65.0m (b) 130.0m (c) 260m (d) 520.0m.
- SOL: $V = fe = 2d/t$, Where d = distance,
 t =time. i.e. $130 \times 2 = 2 \times d / 0.5$, $d = 130 \times 2 \times 0.5 / 2 = 65\text{m}$.
15. In an experiment, lead shot contained in a vertical cupboard cylinder falls through 100cm when the cylinder is inverted. Calculate the rise in temperature caused by 100 such inversions. (Specific heat capacity of lead 130J/kg; $g = 10\text{ms}^{-2}$).
- SOL: Using $mce = \text{work done}$ and $mgh = \text{work done}$. Thus, $mc\theta = mgh$, $c\theta = gh$.
 $\therefore \theta = gh/c = \frac{100 \times 10}{130} = 7.7\text{K}$
16. The quality and pitch of a musical note depend respectively on

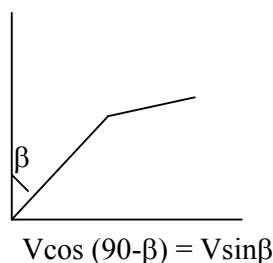
- (a) frequency and harmonics (b) overtones and intensity
(c) intensity and frequency (d) overtones and frequency.

(Quality depends on overtones or harmonics and Pitch depends on frequency. N.B: Loudness depends on intensity).

17. A body is projected with a velocity $V \text{ms}^{-1}$ inclined at angle β to the vertical. Which of the following gives the correct expression for the horizontal component of velocity V_x after time t ?

- (a) $V \cos \beta$ (b) $V \sin \beta$ (c) $V_g \cos \beta$ (d) $V_i \sin \beta$

SOL:



18. A converging lens of radius of curvature 16.00cm, calculate the power of the lens
(a) 0.0625 (b) 0.125 (c) 0.250 (d) 0.500.

SQL: Power = $1/f$,

Recall $f = \frac{\text{radius of curvature}}{2}$,

thus $f = 16/2 = 8$. Power = $1/8 = 0.125$

19. If the velocity of light in air is $3.0 \times 10^8 \text{ms}^{-1}$, find the velocity of light in a medium whose refractive index is 1.5.
(a) 5×10^8 (b) 5.0×10^7 (c) 2.5×10^8 (d) 2.5×10^7

SOL: $D = \text{velocity in air (vacuum)} / \text{velocity in medium. (D = refractive index)}$

$1.5 = (3.0 \times 10^8) \times X = 3 \times 10^8 / 1.52.0 \times 10^8$.

20. When an object is at infinity to the pole of a concave mirror the image is formed at;
(a) principal focus, real and diminished (b) centre of curvature, real and inverted
(c) centre of curvature, virtual and erect (d) principal focus, real and erect.

SQL: Object at infinity, forms an image at the principal locus, real inverted and diminished.

21. A total eclipse of the sun is seen when the observer is in the

- (a) umbra region of the sun's shadow (b) penumbra region of the sun's shadow
(c) umbra region of the moon's shadow (d) penumbra region of the moon's shadow.

(In the eclipse of the sun, the moon is

in between the sun and the earth, therefore the shadow of the moon is cast upon the earth. Total eclipse is experienced in some places and partial eclipse in some other places. The total eclipse region is referred to as the umbra while the partial eclipse region is referred to as the penumbra).

22. A long drawn sound mixing from overlapping reflections together with the original sound which, gives rise to the reflections is called

- (a) beat (b) echo (c) diffraction (d) reverberation.

(Echo is the Sound heard by a person after reflection of the sound caused by him. Reverberation is caused as a result of multiple sound reflections).

23. Which one of the following radiations for is used for examining the freshness or staleness of eggs in the poultry?

- (a) Ultra-violet rays (b) x-rays (c) a-rays (d) Radio waves.

- (The staleness or freshness of an egg can be checked by ultra-violet rays).
24. Which of the following statements is not correct about electro magnetic radiations?
 (a) They are quantized energies (b) They are transverse waves
 (c) They are polarizable (d) They may be deflected in an electric or a magnetic field.
 (Electromagnetic waves are transverse waves, polarizable and also have quantized energies).
25. A simple cell with mercury-amalgamated zinc plate minimizes
 (a) sparking (b) local action (b) back e.m.f (d) polarization.
 SOL.: The two defects of a simple cell are polarization and local action. Polarization can be minimized with the use of a depolarizer such as MnO_2 (Leclanche cell) or CuSO_4 (Daniel cell). While local action can be minimized by coating the zinc surface with mercury-amalgam.
26. Which of the following is not a mixture?
 (a) Seawater (b) Iron (II) Sulphide (c) Petroleum (d) Urine.
 (A mixture is any substance containing two or more components not chemically combined together and can easily be separated into its component parts by physical methods. Seawater, urine and Petroleum are mixtures but Iron (II) Sulphide is a compound. NB: Mixtures have no chemical formula. Of all the chemical samples only Iron (II) Sulphide has a chemical formula (FeS)).
27. Carbon forms two oxides; the ratio of oxygen in both oxides is 1:2. If one mole of carbon separately combines with oxygen, deduce the formula of the oxide.
 (a) CO: CO_2 (b) CO: CO_3 (c) CO_2 : C_2O (d) CO_2 : CO_3 .
 SQL: Oxide 1 Oxide 2
 1 mole C + 1 mole O: 1 mole C + 2 mole O
 CO CO_2
28. If 50cm^3 of CO_2 gas and 50cm^3 of SO_2 gas are measured at S.T.P, calculate the amount of each gas in mole (Molar Gas Volume = 22.4 dm^3).
 (a) 0.023 mol (b) 0.0222 mol (c) 0.0002 mol (d) 0.0022 mol.
 SOL.: From the question, one mol each of CO_2 and SO_2 is used.
 Recall; one mole of any gas occupies a volume of 22.4dm^3 (22400cm^3). Thus if 1 mole of CO_2 occupies 22.4dm^3 x mole of CO_2 will occupy 0.05 dm^3 (converting from cm to dm³). Cross multiply and make X the subject of the formula,
 $X = 1 \times 0.05 / 22.4 = 0.0022\text{ mol}$ of CO_2 . It is the same with the number of moles of SO_2 according to Avogadro's law which states: Equal volume of molecules.
29. How much volume of ethane will be required to produce $1.12\text{m}^3/\text{dm}^3$ of carbon (IV) oxide on combustion in sufficient oxygen?
 $2\text{C}_2\text{H}_6(\text{g}) + 7\text{O}_2(\text{g}) \rightarrow 4\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g})$
 (a) 56dm^3 of ethane (b) 2.24dm^3 of ethane (c) 0.56dm^3 of ethane (d) 5.6dm^3 of ethane.
 SOL: This is volume - volume relationship in stoichiometry between C_2H_6 and CO_2
 NB: 1 mole of any gas occupies a volume of 22.4 dm^3 (MGV).
 If $2 \times 22.4\text{dm}^3$ (2moles) C_2H_6 is required to produce $4 \times 22\text{cm}^3$ (4moles) CO_2 Xdm³ of C_2H_6 will produce 1.12dm^3 of CO_2 Cross multiply and make X the subject the formula,
 $X\text{dm}^3 = 1.12 \times 2 \times 22.4 / 4 \times 22.4 = 0.56\text{dm}^3$.
30. By how much would the volume have increased when 1 dm^3 of ozone were converted to oxygen? $2\text{O}_3 \rightarrow 3\text{O}_2$
 (a) 15dm^3 (b) 10dm^3 (a) 0.5dm^3 (d) 5dm^3 ,
 SOL.: Remember molar gas constant = 22.4dm^3 If $2 \times 22.4\text{dm}^3$ (2moles) of O_3 gave $3 \times 22.4\text{dm}^3$ (3moles) of O_2
 10dm^3 of O_3 will give x dm³ of O_2
 $X = 10 \times 3 \times 22.4 / 2 \times 22.4 = 15\text{dm}^3$

Increase in volume = $V_2 - V_1$,

Where $V_1 = 10\text{dm}^3$ and $V_2 = 15\text{dm}^3$

$= 15\text{dm}^3 - 10\text{dm}^3 = 5\text{dm}^3$.

31. The pressure exerted by a gas is as a result of

- (a) the continuous random motion of its particles
- (b) bombardment of the walls of the container by its molecules
- (c) the collision between the gas molecules
- (d) the elastic nature of the gas.

{According to Pascal, pressure = force exerted per unit area. When gas molecules continuously bombard or hit the walls of their containers, a force is exerted on a given area, which is called pressure. This is experienced in the bubbling cover cork of a kettle during boiling. This vibration is caused by the bombardment of the cork by steam molecules (water vapour)).

32. Which of the following will diffuse at the slowest rate?

- (a) Ammonium (b) Sulphur (IV) oxide (c) Carbon (II) oxide (d) Nitrogen.

SOL: According to Graham's law of diffusion; the rate of diffusion of a gas is inversely proportional to the square root of its density or relative molecular mass at constant temperature and pressure.

i.e. $R \propto 1/d$ or $1/R_{mm}$, thus as rate increases, the "d" or "Rmm decreases

NH_4 $(14+1 \times 4) = 18\text{g Rmm}$,

"18

SO_2 $(32+2 \times 16) = 64\text{g Rmm}$,

"64

CO_2 $(12 + 2 \times 16) = 44\text{g Rmm}$,

"44

N_2 $(14 \times 2) = 28\text{g Rmm}$,

"28

Therefore, SO_2 with the highest molar mass diffuses at the slowest rate.

33. Element P has an electronic configuration of 2, 8, 6. Element R has an electronic configuration of 2, 8, 8, 1. What is likely to be formed by P and R?

- (a) A covalent compound PR (b) An ionic compound PR_2
- (c) An ionic compound PR (d) An ionic compound P_2R

SOL: Element P Element R 2,8,6 2,8,8,1

Element P is a group 6 non-metal while element B is a group I metal. Such a combination is ionic or electrovalent.

Using Radicals simultaneous method for electrovalent combination

No. of deficient electrons

(P)=2(1)

No. of excess electrons

(R)=1 (2)

Make No. of deficient electrons = No.

of excess electrons i.e. eqn 1 = eqn 2

$\text{P}=2 \times 1, \text{R}=1 \times 2$.

Thus make the multiple of P and R a subscript for them. i.e. $\text{P}_1 \text{R}_2 = \text{PR}_2$

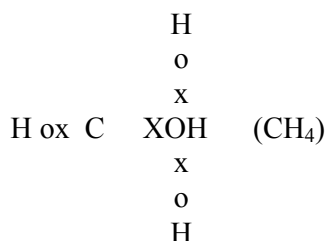
34. Which is the atomic structure of phosphorus with mass No. of 31?

- (a) Protons = 15, Neutrons = 16, Electrons = 15. (b) Proton = 15, Neutrons = 16, Electrons = 16,
- (c) Protons = 16, Neutrons = 15, Electrons = 15. (d) Protons = 16, Neutrons = 15, Electrons = 16.

- SOL: Since the atom is neutral, No. of protons = No. of electrons. Phosphorus has 15 protons and 15 electrons. But mass No. = Protons + Neutrons
 $\therefore$ Neutrons = mass No. - Protons = 31-15 = 16. Thus protons = 15, neutrons = 16, electrons = 15.
35. Which metal has the least tendency to form positive ions?
 (a) Iron (b) Aluminum (c) Sodium (d) Calcium
 (In the periodic table of elements, ionization energy increases from left to right progressively but decreases down the group. Thus, the higher the number of valence electrons, the higher their ionization energies (decrease in tendency to form positive ions)
 Fe has 2 valence electron, Na has 1 valence electron, Ca has 2 valence electrons but Al has 3 valence electrons).
36. Natural water includes the following except
 (a) rain water (b) spring water (c) pure water (d) lake water.
 SOL: Of all the options, only pure water is not a source of natural water. Sources of natural water includes rain or precipitation, spring, stream, well, lake, pond, sea etc. Pure water as used in the option means treated water, which is not actually pure, following the criteria for purity.
37. Which of the following methods cannot be used to remove permanent hardness from water?
 (a) Addition of washing soda (b) Addition of caustic soda
 (c) Permutit method (d) Adding alum to water.
 {Permanent hardness is caused by the presence of Ca and Mg ions in water either as chlorides, carbonates, hydrogen carbonates, or sulphates. These ions can be substituted using soluble metals like sodium containing compounds such as washing soda ($\text{Na}_2\text{CO}_3 \cdot 10 \text{H}_2\text{O}$), Caustic soda (NaOH), and Permutit or Zeolite (NaAlSiO_3) NB: Alum is a double salt which conforms to the general formulae $\text{M}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$. Where M^+ can be Na^+ , K^+ or NH_4^+ and M^{3+} can be Al^{3+} , Cr^{3+} or Fe^{3+}
38. If the solubility of sodium tetraoxosulphate (IV) at 30°C is 18g per 100g. How much is this in gram per kilogram?
 (a) 18kg per 1000g (b) 180g per 100g (c) 180g per 1000g (d) 180g per 1kg.
 SOL: $18\text{g per } 100\text{g} = 18\text{g per } 100/1000\text{kg}$
 (converting from g to kg) = $18\text{g per } 0.1\text{kg} = 180\text{g per } 1\text{kg}$.
39. The following are examples of colloid except
 (a) milk (b) starch in water (c) aerosol (d) ammonium chloride solution.
 (A colloidal solution is one in which there are suspended particles in a medium capable of reflecting some amount of light when viewed through a light source. This phenomenon is called Tyndal's effect. Milk, starch and aerosol are colloids except ammonium chloride solution which has no suspended particles, thus forming a clear solution). NB: Colloidal solutions are also called false solutions.
40. The pH of the solution M, N, O and P are 4, 6, 8 and 10 respectively. Therefore,
 (a) none of the solution is acidic (b) the H of O is made neutral by adding water
 (c) the most acidic solution is P (d) M is the most acidic solution.
 (According to Sorensen's equation, $\text{pH} = -\log_{10} \text{H}^+ = \log_{10} 1/\text{H}^+$
 Therefore, a solution with a lower H and a higher H^+ concentration is more acidic than that with a higher P and a lower H^+ concentration).
41. Sodium chloride may be obtained from brine by
 (a) decantation (b) distillation (c) evaporation (d) sublimation.
 {Brine is a concentrated solution of table or common salt (NaCl) and water (H_2O). While NaCl alone can be gotten from brine by evaporation to dryness; both NaCl and H_2O can be collected separated by distillation, due to large difference in their boiling points with NaCl higher in boiling point than H_2O }.
42. Oil Spillage in ponds and creeks can be cleared by
 (a) burning off the oil layer (b) spraying with detergent

- (c) spraying with common salt (d) spraying with oil
- (Oil is an immiscible solvent with water, forming the upper layer due to its lower density. This spreads over the surface, cutting off the oxygen supply to the aquatic bodies from the atmosphere (reducing dissolved oxygen). Detergent having hydrophilic head and hydrophobic tail interlink these two immiscible liquids and dissolves the oil).
43. Which of the following is a chemical compound?
- (a) Soap (b) Milk (c) Urine (d) Gold.
- {A chemical compound contains atoms of a particular element and that of other elements combining together in a means and have chemical formulae not symbols which is for elements.
- Soap = $C_{17}H_{35}Na$ (compound). Milk = No formulae (mixture). Urine = No formulae (mixture). Gold = Au (element)).
44. Crystallization is a separation method used
- (a) where purity of the product is important
- (b) where beauty of the product is important
- (c) where the product is a solid
- (d) where the products cannot be destroyed by heat.
- {When salt solution containing impurity is heated to a particular temperature, it is allowed to cool to a certain temperature where the pure crystals come out of the solution leaving behind the impurities in solution. Crystallization can be induced by scratching and salting out (adding of crystals of the same salt)}.
- NB: It is used where a salt can be easily decomposed by heat.
45. Which hydroxide dissolves in water to form an alkaline solution?
- (a) Aluminum hydroxide (b) Calcium hydroxide (c) Copper hydroxide (d) iron hydroxide.
- (A soluble oxide or hydroxide is called an alkali. Dissolves in water to form an alkaline solution. Other soluble hydroxides (alkalis) include NaOH, KOH, LiOH, $NH_3 \cdot H_2O$, etc).
46. Starch molecules can be broken down into smaller molecules by heating with a dilute acid. What is this type of reaction called?
- (a) Cracking (b) Hydrolysis (c) Oxidation (d) Reduction.
- {Cracking is the process of breaking down larger molecular petroleum fractions into smaller molecular units of desired products; There are three types of cracking; thermal cracking, hydro cracking and catalytic cracking. Oxidation is the addition of oxygen, removal of hydrogen, addition of electronegative or removal of hydrogen, addition of electronegative or removal of electrons from an atom or a compound. Reduction is the reverse process of oxidation. Hydrolysis means the breakdown of higher molecules (macromolecules) by addition of acid and water. Sucrose hydrolysis yields two invert sugars; glucose and fructose i.e. an aldohexose and a ketohexose}.
47. Which of the following cannot be used as an oxidizing agent?
- (a) $K_2Cr_2O_7$ (b) CO_2 (c) H_2S (d) HNO_3
- (See solution to question 46 $K_2Cr_2O_7$, CO_2 and HNO_3 have the ability to donate oxygen, except H_2S which is a strong reducing agent).
48. What is the number of pairs of shared electrons in a methane molecule?
- (a) 1 (b) 29 (c) 3 (d) 4

SOL:



In this molecule, we have a total of 8 electrons according to the Lewis octet electronic structure i.e 8 shared electrons but 4 pairs of shared electrons.

49. The percentage by volume of nitrogen in the air is high because

(a) nitrogen is relatively inactive (b) nitrogen supports life
(c) nitrogen prevents corrosion of metals (d) nitrogen increases the rate of combustion.

(Options b and c are the functions of Nitrogen. It also decreases the rate of combustion and corrosion. It has a fairly high percentage in due to its inactive nature, which is almost inert (unreactive).

50. What quantity of copper will be deposited by the same quantity of electricity that deposited 18g of aluminum
(a) 64 (b) 32 (c) 16 (d) 8.

SOL: First, calculate the quantity of electricity, needed for the decomposition of 18g of aluminum.

$\text{Al}^{3+} + 3\text{e}^-$ (3 faradays of

electricity. But 1 faraday = 96500

Coulombs)

27g 3 x 96500 Coulombs

18g X Coulombs, $X = 18 \times 3 \times$

$96500/27 = 1\,93000$ Coulombs.

Finally, calculate the quantity of copper

deposited by 1 9300Coulombs.

$\text{Cu}^{2+} + 2\text{e}^-$ (2 faradays of electricity)

64g 2 x 96500 Coulombs

Xg 19300Coulombs, $X = 19300 \times 64/2$

$\times 96500 = 64\text{g}$.

51. When two genes for the same character are contained in the same individual, the character that shows is known as

(a) important character (b) dominant character (c) superior character (d) controlling character.

52. In Nigeria, a tropical rainforest can be found in

(a) Sokoto (b) Kaduna (c) Abuja (d) Abia.

(Tropical rainforest is associated with dense and thick vegetations arranged in strata Or layers, with evergreen leaves. They are mostly found in the southern, eastern and western parts of Nigeria. Of all the options, only Abia is an eastern state).

53. The sum total of the biotic and abiotic factors that affect living things is referred to as

(a) environment (b) lithosphere (c) hydrosphere (d) atmosphere.

(Environment is the sum total of both living (biotic) and nonliving (abiotic) factors that affect living organisms. Lithosphere is the solid layer of the earth crust. Hydrosphere is the liquid part of the cosmos. Atmosphere is the gaseous envelope that surrounds the earth, while the living component of earth is the Biosphere or Ecosphere).

54. Hygrometer is an ecological instrument that measures

(a) rainfall (b) humidity (c) temperature (d) light.

SOL: ECOLOGICAL FACTORS

MEASURING INSTRUMENTS

Rainfall

Rain gauge

Humidity

Hygrometer

Temperature

Thermometer

- Light
Photometer
Turbidity
Secchi disc.
Wind speed
Anemometer
Wind direction
Wind vane.
55. *Candida vagirfalis* is a
(a) bacterium (b) fungus (c) virus (d) protozoan.
56. Which one of the following is airborne?
(a) Malaria (b) Yellow fever (c) Cholera (d) Tuberculosis
(Malaria is caused by *Plasmodium malarae* from mosquito (female anopheles) bite. Yellow fever is caused by arbovirus from the of infested *Aedes* mosquito. Cholera is contracted through drinking of contaminated water. It is caused by *Vibrocholerae*. Tuberculosis is airborne; it is caused by *Mycobacterium tuberculosis*).
57. All of these are vertebrates except
(a) lizard (b) rat (c) star fish (d) tilapia
(Lizard belongs to the class Reptilia. Rat belongs to the class Mammalia. Starfish belongs to the class Reptilia. Rat belongs to the class Mammalia. Starfish belongs to the phylum Echinodermata. Tilapia belongs to the class Pisces. One major characteristic of all vertebrates is the presence of backbone (vertebrae column). Thus lizard, rat and tilapia are vertebrates, while starfish is an invertebrate).
58. A box was left in the lawn for two days. When the box was removed, the grass under had turned yellow due to lack of
(a) carbon dioxide (b) light (c) oxygen (d) water.
(While the grass was covered, photosynthesis ceased to occur in that area due to the absence of light responsible for photolysis of water to produce H^+ and OH in the light phase which enters into the dark phase where CO_2 is reduced by $[t]$ to form CH_2O (starch). It turned yellow because of lack of nutrients resulting from the absence of sunlight, which is the initiator).
59. Which of the following deaminates excess amino acid?
(a) Duodenum (b) Ileum (c) Liver (d) Kidney
Deamination means the breakdown of amino acids into urea and keto acid. The urea is excreted, while the keto acids perform the functions peculiar to the glucose. It is one function of the liver apart from its storage (glucose. It is one function of the liver apart from its storage (glucose, iron etc) and secretory (bile for fat emulsification)).
60. In which of the following floral parts does meiosis occur?
(a) Anther (b) Petal (c) Receptacle (d) Style
(Meiosis occurs after fertilization of the ovules in the ovary by the pollen grains from the anther. The fertilized ovules divide within the style (a) region above the ovary). At maturity, ovules = seeds and ovaries = fruits).
61. When an organism moves its whole body towards a stimulus, the organism is said to exhibit
(a) tropic movement (b) trophic movement (c) tactic movement different ways;
these are Tropism which involves movement of part(s) of the organism's body to the direction or away from a directional stimulus. E.g. phototropism (light), geotropism (gravity), thigmotropism (touch), chemotropism (chemical), Tactism, which involves the movement of the whole body of an organism in, responds to a directional stimulus. e.g. Chemotactism (chemical) thermotactism (temperature) thigmotactism (touch) and Nastism (Sleep movement) where part(s) of an organism responds to a diffused stimulus}).
62. A stable self sustaining environment produced by an interaction between the biotic and abiotic components is best described as

- (a) niche (b) a community (c) an habitat (d) an ecosystem.
(Niche is the position occupied by an organism together with its functional role in the habitat (residence). Population is a collection of plants or animals or animals of the same specie living together in a particular environment and free to interbreed. Community is the population of different species of organisms in a particular environment. While an ecosystem is an interaction between living (biotic) and nonliving (abiotic) components of the environment).
63. Which of the following is not present in the nucleus of a cell?
(a) Chromosomes (b) Nucleus (c) Genes (d) Mitochondria
(The living component of the cell is the protoplasm = nucleus + cytoplasm. Hereditary materials such as chromosomes and genes are found in the nucleus, while the cytoplasm is a medium for cellular metabolism (biochemical reactions) where organelles like mitochondria, endoplasmic reticulum, Golgi bodies etc are found).
64. The release of useful substances from the cells of an organism is called
(a) excretion (b) evacuation (c) metabolism (d) secretion.
(Useful substances like hormones and digestive enzymes are secreted, not excreted by ductless (endocrine) and ductful (exocrine) glands for a particular action in the body. While excretion is the removal of waste product of metabolism from a cell).
65. The head of the femur articulates with the pelvis at the
(a) glenoid cavity (b) olecranon process (c) acetabulum (d) coracoid process.
(The femur (thigh bone), which is the longest bone in the body, articulates with the pelvis at the glenoid cavity. NB: The smallest bones in the body are the ear ossicles viz; hammer (malleus), anvil (incus) and stirrup (stapes)).
66. The mammalian cervical vertebrae invariably numbers
(a) 4 (b) 7 (c) 12 (d) 15
(The human vertebrae column is arranged thus: Cervical = 7 bones. Thoracic = 12 bones. Lumbar = 5 bones. Sacra = 5 bones. Caudal 4 bones. Making a total of 33 bones).
67. What mechanism is responsible for the rise of water to the top of all trees?
(a) Transportation pulls (b) Root pressure (c) Osmosis (d) Capillarity.
(The xylem vessel transports water due to root pressure, transpiration pull and capillary action. Root pressure and capillary action alone can carry water to a short distance, which is now assisted by transpiration pull, developed when water moves up the stem, due to loss of water from the stomata of leaves by the process of transpiration, thus enhancing long distance transportation in higher plants).
68. Antibodies in mammalian blood are formed by
(a) platelets (b) white blood cells (c) red blood cells (d) liver.
(Platelets (Thrombocytes) help in blood clotting by the formation of fibrin network structure on wounds formed when thrombin combines with fibrinogen, preventing blood attack from toxins secreted by disease causing microorganisms, by the formation of antibodies. They are often referred to as the body soldiers for this function of defense. Red blood cells (Erythrocytes) transport food and gases from the digestive system and the lungs to the body cells for utilization).
69. Which of the following liquids supplies cells in the tissues of a mammal with oxygen and nutrients?
(a) Blood (b) Plasma (c) Serum (d) Lymph
(See solution to question 68).
70. The cockroach and grasshoppers have mouthparts adapted for
(a) sucking (b) piercing and sucking (c) biting and chewing (d) biting and lapping.
(Grasshopper and cockroach have strong manipulate mouthparts for biting and chewing food particles).

71. The sponging and lapping mouthparts occur in
(a) housefly (b) butterfly (c) cockroach (d) mosquito.
(The housefly has proboscis from modified labium. The proboscis is spong like and consists of two halves called labella. The proboscis is used in feeding by digesting any solid food extra cellularly before sucking it in).
72. Which of the following is not an example of sex-linked character?
(a) Baldness (b) Colour blindness (c) haemophilia (d) Height
(Baldness, colour blindness and haemophilia are sex linked because they cannot be transferred to the next generation through the XY chromosomes but through XX chromosomes where they are recessive traits; except height which is not sex linked).
73. Which of these is a vestigial structure?
(a) Appendix (b) Caecum
(c) Pancreas (d) Sacculus rotundus
(Vestigial organs or structures are structures that not useful to some animals but useful to others. e.g. the appendix in man is non-functional. But it functions as a storage site for microbes (aecum) for cellulose digestion in herbivores).
74. Which of the following air pollutants causes acid rain?
(a) Sulphur dioxide (b) lead oxide (c) carbon dioxide (d) hydrogen dioxide
(Acid rain is experienced when water reacts with acid anhydrides in the atmosphere from air pollution, e.g. CO₂, SO₂, NO₂, etc, forming an acid. Though options A, C, D, are correct but acid rain is mostly associated with SO₂, because it forms a pH lower than 5, 6 when dissolved in water).
75. Which of the following does not have a well- developed tissue?
(a) Moss (b) Fern
(c) Whispering tree (d) Maize.
(Of all the options, B, C and D have tissues like true root, stems and leaves. Moss is a bryophyte that lacks true roots, stems, and leaves).

ANSWERS

- 1.D 2.C 3.C 4.A 5.nIL
6.A 7.n 8.A 9.n 10.B
11.C 12.C 13.D 14.A 15.D
16.D 17.B 18.B 19.n 20.A
21.C 22.D 23.A 24.D 25.B
26.B 27.A 28.D 29.C 30.D
31.B 32.B 33.B 34.A 35.B
36.C 37.D 38.D 39.D 40.D
41.C 42.B 43.A 44.A 45.B
46.B 47.C 48.D 49.A 50.A
51.B 52.D 53.A 54.B 55.B
56.D 57.C 58.B 59.C 60.D
61.n 62.D 63.D 64.D 65.C
66.D 67.A 68.B 69.A 70.C
71.A 72.D 73.A 74.A 75.A

POST UNIVERSITIES MATRICULATION SCREENING TEST 2006

SECTION A: USE OF ENGLISH

TIME ALLOWED: 1 HOUR

Reading passage I carefully and answer the questions that follow.

When survey all known human societies, we find everywhere a form of family, some set of permanent arrangement by which the males assist the females in caring for the children while they are young. The distinctive human aspect of the enterprise lies not in the protection the male affords the females and the young, this we share with the primates, nor does it lie in the lordly possessiveness of the male over the female for whose favour he contends with other males-this too we share with the primates. Its discipline lies instead in the nurturing behaviour of the male who among human beings everywhere helps provide foods for the women and children.

Among our structurally closest analogues-the primates, the males do not feed the females, he may fight to protect her or to possess her, but he does not nurture her. She has to fend for herself.

Inventions were made under which males started nurturing females and their young.

In every known human society, everywhere in the world, the young male learns that when he grows up, one of the things he must do in order to be a full member of the society is to provide food for some females and her young. Even in very simple societies, a large number of men may escape the burden of feeding females and young by entering monasteries and feeding each other or by entering some profession that their societies will classify as giving them a right to be fed, like the Army and the Navy or the Buddhist order of Burma. But in spite of such exceptions, every known human resists firmly on the learned nurturing behaviour of men.

1. Man differs from other animals because
 - (a) he forms a family
 - (b) cares for the children
 - (c) he cares for women
 - (d) he provides food for women
2. The "lordly possessiveness" of the male over the female means that the male
 - (a) shows mastery over the female
 - (b) is more aggressive towards the females
 - (c) claims right over the female
 - (d) protects the female.
3. "Somewhere at the dawn of human history" refers to
 - (a) the beginning of Creation
 - (b) the creation of man
 - (c) the time when man was monkey
 - (d) the beginning of civilization
4. Man takes it upon himself to feed the female except when
 - (a) he takes up the gun to fight
 - (b) he becomes religious
 - (c) he forms a religion
 - (d) he enters some profession.
5. According to the passage, a common feature of all human communities is
 - (a) the family
 - (b) caring mother
 - (c) aggressive father
 - (d) nurtured women and children

Read Passage 2 carefully and answer the question that follows.

From ancient times, people commonly believed that life arose repeatedly by spontaneous generation from non-living materials in addition to parental reproduction. For example, frogs appeared to arise from damp earth, mice from putrefied matter, insects from dew and maggot from decaying meat. Warmth, moisture, sunlight and even starlight often were mentioned as factors that encouraged spontaneous generation of living organisms.

Among the accounts of early efforts to synthesize organism in the laboratory is a recipe for making mice, given by the Belgian plant nutritionist Jean Baptiste van Helmont (1648). "If you press a piece of underwear soiled with sweat together with some wheat in an open jar, after about twenty one days, the odour changes and (he ferment changes the wheat into mice. Hut what is more remarkable is that the mice which came out of (lie wheat and underwear were not small mice, not even miniature adults or aborted mice but adult mice emerged.

In 1961, the great French scientist Louis Pasteur convinced scientists that living organisms cannot arise spontaneously from non-living matter. In his famous experiment, Pasteur introduced fermentable materials into a flask with a long S-shaped neck that was open to air. The flask and its contents were boiled for a long time to kill any microorganisms that might be present. Afterwards, the flask was cooled and left undisturbed. No fermentation occurred because all organisms that entered the open end were deposited in (he open neck and did not reach the fermentable material. When (he neck of (he flask was removed, microorganisms in the air promptly entered the fermentable material and proliferated. Pasteur concluded that life could not originate in the absence of previously existing organisms and their reproductive elements, such as eggs and spores. Announcing his results to the French Academy, Pasteur proclaimed, "Never will the doctrine of spontaneous generation arise from this mortal blow". All living organisms share a common ancestor, most likely a population of colonial microorganisms that lived almost four billion years ago. This common ancestor was itself a product of a long period of prebiotic assembly of non-living matter, including organic molecules and water, to form self-replicating units. All living organisms retain a fundamental chemical composition inherited from the ancient common ancestor.

6. Throughout the passage, the word "spontaneous" can best be taken to mean
 - (a) without reproductive element
 - (b) in a medium
 - (c) unthinking
 - (d) spirited.
7. In Pasteur's experiment, why was the neck of the flask removed?
 - (a) To allow air to escape
 - (b) To provide access to microorganisms
 - (c) To kill any micro organisms that may be present
 - (d) To prevent the heating of the flask.
8. In the third paragraph, the word "mortal" most nearly means
 - (a) human
 - (b) impermanent
 - (c) fatal
 - (d) earthly.
9. The expression "when the neck of the flask was removed" found in the third paragraph can be described as
 - (a) adverbial clause of time
 - (b) adjectival clause
 - (c) noun clause
 - (d) adverbial phrase of time.
10. In what sense is the use of the word "water" in the last paragraph ironic?
 - (a) It is a medium in which many living organisms thrive
 - (b) It has a relatively low boiling point
 - (c) Pasteur did not mention it specifically in his description of his experiment
 - (d) It was not one of the ingredients in Helmont's recipe.

Sonic secrets of comprehension

- Read and carefully digest the text, highlighting key words.

- Read the passage trying to figure out the key words already noted.
- Read the questions ones more and provide Her answers correctly.

NB: All these must be done with speed and accuracy because you have a limited time.

Read the following passage and select the best option that fills the corresponding gap in the text.

By the 28 week, that is the end of the 7th month, the foetus moves around vigorously with (he—12—and its heartbeat can be heard by the—13—Its length is 38cm (15 inches) and its weight 1000grams. Its body is thin, its skin reddish and Covered with n—14— coating of a creamy, waxy substance, which is manufactured by small glands in the skin. It can open its eyes and the —15— covering (the pupils of the—16—has gone. If it is — 17— at this stage, it can —18— but with difficulty. It can also cry —19—, but —20—it's sleep energetically.

- | | (a) | (b) | (c) | (d) |
|-----|-----------|--------|-----------|---------|
| 11. | Day | Week | Month | Year |
| 12. | Stomach | Uterus | Bladder | Belly |
| 13. | Audience | Women | Crowd | Doctor |
| 14. | Defensive | Kickin | Plastic | Protect |
| 15. | Membrane | Hand | Glass | Rubber |
| 16. | Eyes | V | Class | Mouth |
| 17. | Left | Sent | Discharge | Born |
| 18. | Breathe | See | V | Hear |
| 19. | Angrily | Weakly | Loudly | V |
| 20. | Walk | Wove | V | Wraps |

In questions 21-25, choose the option that has the opposite meaning to the word or group of words underlined in the sentences to fill the missing word(s).

21. One would have expected the students to take a consultation with the school authorities instead of reporting to....
(a) mediation (b) confrontation (c) arbitration (d) exultation
(“Consultation” is “the act of discussing something with somebody or group of people before making a decision about it”. Its antonym is “confrontation”).
22. It is surprising that one identical is an alcoholic but the other is
(a) teetotaler (b) dipsomaniac (c) agoraphobic (d) saboteur.
(The antonym of “alcoholic” is “teetotaler”, which means “a person who does not drink alcohol”).
23. While the man is parsimonious, his wife is obviously....
(a) ingenious (b) curious (c) judicious (d) extravagant.
(The word “parsimonious” means “extremely unwilling to spend money”. Its antonym from the above options is “extravagant” which means “spending a lot of money”).
24. The lackadaisical student failed the test, but theone passed it well
(a) enthusiastic (b) listless
(c) classical (d) obtuse.
(“Lackadaisical” refers to “lack of zeal, zest or willingness”. Its antonym is “enthusiastic” which has to do with “zeal”).
25. While a sinner will incur the wrath of God, a saint will enjoy the of God.
(a) anger (b) anointing (c) favour (d) pleasure
(“Wrath” means “extreme anger”).

In question 26-35, from the words lettered A - D, choose the one that best complete the sentence.

26. The rich tnnn ... money lo his wives and children every Monday,
(a) disperses (b) dispensed (c) discuss (d) dispenses.
("Dispense" means to give out something to people).
27. If I had known that the man was not a genuine pastor, J would not have ... too much confidence in him.
(a) imposed (b) reposed (c) inputted (d) disposed
28. 'The poor woman has ... herself to fate,
(a) consigned (b) concerned
(c) resigned (d) reposed.
29. The clergy ... everyday for the success of (he present government,
(a) are praying (b) is praying (c) pray (d) prays.
30. The young girl,mind her language,
(a) have better (b) had better (c) have best (d) had best.
31. Many a secondary school student afraid of English Language examination nowadays.
(a) are (b) is (c) was (d) were.
32. The man with his three wives and seventeen children — just traveled abroad,
(a) is (b) are (c) has (d) have.
33. The lecturer does not ... his real age.
(a) seems (b) look (c) appear (d) resemble.
34. During the recent communal crisis, the place of the monarch was ... with charms and amulets
(a) decorated (b) fortified (c) embellished (d) resemble
(To "fortify" means to "strengthen a place against attack").
35. I ... your father to stop going out with politicians; than I changed my mind,
(a) had told (b) have told (c) was going to tell (d) have being telling.

In questions 36 - 40, a list of options is given below every sentence; choose from the options the one nearest in meaning to the word or phrase underlined in the sentence

36. Are they expecting celestial blessings?
(a) terrestrial (b) heavenly (c) religious (d) sacrificial
('Celestial' has to do with "sky or heaven").
37. The indigent is my younger brother's friend
(a) indolent (b) diligent (c) efficient (d) needy.
("Indigent" means "very poor or needy").
38. Adults seldom visit the new amusement center,
(a) frequently (b) rarely (c) usually (d) reluctantly.
("Seldom" means "rarely", "hardly", "less often").
39. Really life is ephemeral
(a) enjoyable (b) difficult (c) transient (d) permanent.
(ephemeral has to do with lasting or used for only a short period of time. Its synonym is transient).
40. I want to tell you that Felix has an eclectic mind
(a) versatile (c) critical (b) shocking (d) narrow
(The adjective "eclectic" means "not following one style or set of ideas but choosing from or using a wide variety". Its synonym above is "versatile" which means "able to do many things").

ANSWERS

- 1.D 2.D 3.D 4.D 5.A
 6.A 7.B 8.D 9.D 10.A
 21.D 22.A 23.A 24.A 25.C
 26.D 27.A 28.C 29.D 30.B
 31.B 32.C 33.A 34.B 35.C
 36. B 37.D 38.D 39.C 40.B

SECTION B: GENERAL PAPER

CHEMISTRY

PHYSICS

BIOLOGY

1. A sample of orange juice has a P of 3.80. What is the molar concentration of hydrogen ion in the juice?

(a) 1.58×10^4 (b) 1.58×10^2 (c) 1.58×10^3 (d) 1.58×10^4

SOL: pH =

$H = \text{Log}_{10}(H^+)$ (according to Sorensen).

3.80 = $-\log_{10}(H^+)$

$\text{Log}_{10}(H) = -3.80$

$(H^+) = \text{antilog } 3.80$ or

$(H^+) = 1/\text{antilog } 3.80$

$= 1.58 \times 10^{-4}$ - 1.58×10^{-4} .

2. Which of the following is not acidic?

(a) Aluminum chloride (b) Zinc chloride (c) Copper (II) tetraoxosulphate (VI) (d) Silver chloride

(Acidic salts are formed from strong acids and weak bases, basic salts from strong acids and weak acids, while normal salts are formed from either strong acids and strong bases or weak acids and weak bases. Many a student may look at the whole salts as normal due to the absence of H or OH; yes they are, but on hydrolysis (in solution), they exhibit either acidic, basic or neutral properties. Bearing this in mind, AlCl_3 , CuSO_4 and AgCl are acidic except ZnCl_2 which is neutral on hydrolysis).

3. Calculate the current in amperes required

to produce 18.0g of Aluminum in 1.50

hours,

(a) 33.65A (b) 35.74A (c) 37.83A (d) 39.35A

(Al = 27, F = 96500C).

SOL: $\text{Al}^{3+} + 3e^- \rightarrow \text{Al}$ (3 Faradays, but

1 faraday - 96500C) 27g required 3

$\times 965000$, 18.0g will require Q Coulombs, $Q = 18 \times 3 \times 96500/27 = 1930000$ coulombs.

But Q (Charge) = It. (Covert time in hours to time in seconds).

- 193000 - 1.50 x 3600 x I. Thus, I =
193000/1.50 x 3600 = 35 74Amperes.
4. Calculate the oxidation number of chlorine in Cl_2O_7 ;
(a) 17 (b) -2 (c) 5 (d) 11
SOL: Oxygen has an oxidation number -2, Chlorine has oxidation number X
 $\text{Cl}_2\text{O}_7 = 2 \times 4 + (-2 \times 7) = 0$ (having no net charge).
 $\text{Cl}_2\text{O}_7 = 2X - 14$ $2X - 14 = 0$, $2X = 14$, $X = +7$
Thus $2X = +14$; $X = +7$
5. The two nuclei of the hydrogen molecule are held together by
(a) mutual attraction (b) mutual sharing of electrons
(c) a dative covalent bond (d) the two electrons having the same spin,
{Hydrogen molecule is formed when the 1s electron from each atom of hydrogen goes into mutual relationship (share electrons amongst each other) to attain a stable duplet structure of helium}.
6. Which of the following statements about catalysis is correct? (i) A small amount of catalyst often affects the rate of chemical reactions for a long time (ii) A catalyst is always chemically unchanged at the end of the reaction (iii) The effect of a catalyst is often enhanced by adding promoters (iv) A catalyst is always physically unchanged at the end of the reaction (v) A catalyst speeds up the rate of chemical reactions.
(a) i, ii, iii, iv, v (b) i, ii, iii only (c) ii, iii, iv only (d) ii and iv only.
(The only wrong and incomplete statements are (iv) and (v). Catalysts are physically changed in terms of colour, texture, odour, taste at the end of the reaction and a catalyst does not only speed up the rate of chemical reaction but also lowers chemical rate thus it alters chemical reaction rate).
7. When $\text{K}_2\text{Cr}_2\text{O}_7$ dissolves in water, the following equilibrium is established $\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{CrO}_4^{2-}(\text{aq}) + 2\text{H}^+(\text{aq})$. State the colour observed on adding a few drops of dilute H_2S in the system.
(a) Orange (b) Pink (c) Yellow (d) Colourless
8. Which of the following statements is not true for Hydrogen Sulphide?
(a) It is a liquid at room temperature (b) it is a covalent compound
(c) It is a weak dibasic acid in aqueous solution
(d) It is a much stronger reducing agent than water.
(At room temperature, H_2S exists as a gas and not a liquid).
9. Which of the following are products of acid hydrolysis of sucrose?
(a) Fructose and Maltose (b) Glucose and Cellulose
(c) Maltose and Cellulose (d) Glucose and Fructose.
SOL: Sucrose, a disaccharide is broken down into two monosaccharides; Fructose and Glucose on hydrolysis with acidic or enzymatic breakdown.
 $\text{C}_{12}\text{H}_{22}\text{O}_{11} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + \text{C}_6\text{H}_{12}\text{O}_6$
Sucrose glucose fructose
The major difference between glucose and fructose is that while glucose is an aldohexose, fructose is a ketohexose.
10. What is the concentration in mol/dm³ a solution containing 0.28g potassium hydroxide per 100cm³ solution? [K = 39, H = 1, O = 16]
(a) 0.028 (b) 0.04 (c) 0.05 (d) 0.07
SOL: Concentration in mol/dm³ = Molarity.
Molarity = No. of moles of solutes x
1000 / Volume of solution (cm³)
No. of moles = Mass / Molar mass = 0.28 /

39 = 0.0071 moles, Volume = 100cm

Thus, Molarity = $0.0071 \times 1000/100 =$

0.07mol/dm³

11. Determine the distance between two points P (8,5,3) and Q (2,1,-2).

(a) 3.87 (b) 3.32 (c) 15.00 (d) 8.77

SOL: Using $Y = \sqrt{(a-a)^2 + (b-b)^2 + (c-c)^2}$

abc

abc

P(8,5,3)

Q(2,1,2)

$Y = \sqrt{(8-2)^2 + (5-1)^2 + (3-2)^2}$, $Y = \sqrt{6^2 + 4^2 + 1^2}$

$Y = \sqrt{77} = 8.7749 = 8.77$

12. One person walk south-east at 4m/s and another heads north at 3m/s. How far apart are they in 10 seconds if they started at the same point simultaneously?

(a) 3m (b) 28.3m (c) 64.7m (d) 41 9.9m

$F_a = 3\cos 90^\circ + 3\sin 90^\circ$, $F_b = 4\cos 315^\circ + 4\sin 315^\circ$

$F_a \pm F_b = 2.82i + 0.172j + 2.822i + 0.1722j = 2.83m$. $v = d/t$,

$D = vt$, $= 2.83 \times 10 = 28.3$.

13. The centre of gravity of a rectangular object is at

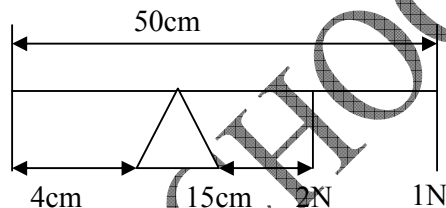
(a) The intersection of the median (b) the middle of the rectangle

(c) the intersection of the diagonals (d) any of the vertices

SOL: The centre of gravity is determined by the intersection of the diagonals.

14. A uniform rule of weight 1.0N is pivoted at the 4cm mark. A meter rule of 2N is hung from the 15cm mark, where must a weight of 2N be placed to balance the rule

(a) 60cm (b) 45cm (c) 55cm (d) 35cm



To determine where 2N will be place for the meter rule to be in equilibrium.

Total clockwise moment = Total anticlockwise moment

i.e, $1 \times (50-4) + 2 \times (15-4) = 2x$

$1 \times 46 + 2 \times 11 = 2x$, $46 + 22 = 2x$,

Therefore $2x = 68$, $x = 68/2 = 34$

15. Two mirrors are inclined at an angle of 60 to each other; determine the number of images in the two mirrors?

(a) 6 (b) 2 (c) 5 (d) 11.;

Using the formula $(360/\theta) - 1$, where θ is the angle between the mirrors.

$(360/60) - 1 = 6 - 1 = 5$.

16. When an oil drop is placed gently on a clean water surface of 100cm

(a) the drop spreads until the film is 5 molecules thick

(b) the drop spreads to fill exactly an area of 100cm

(c) the volume of the drop decreases as it spreads

(d) the oil film formed is not less than one molecule thick,

- (This oil drop experiment was used to determine the diameter of a molecule as ranging $x > 10 \times 10^{-7}$).
17. When illuminated by a light beam, a smoke particle in oxygen can be seen moving in different direction all the time when viewed in a microscope. This is because
- the light beam intensity is not constant
 - the smoke particle is affected by gravity
 - oxygen molecules hit the smoke particles at different directions
 - the smoke particle has a high speed in oxygen.
- (Brownian theory postulates that when smoke particles are observed in air by a microscope, there is an irregular motion of the same put titles as a result of continuous bombardment of the smoke particles by molecules of air around it. This theory can also he illustrated by the bombardment of dust particles by air when viewed through a source of light or bombardment of pollen grains by water molecules when viewed or observed under a microscope)
18. A siren having a ring of 200 holes make 132 revolutions per minutes, a jet of air is directed on to the set of holes; calculate the frequency of the note produced. (Speed of sound in air – 350in/s)
- 26.011
 - 1.5211
 - <100M/
 - 8.66Hz
- SOL: 200 holes is completed in 132 rev/nun I holeis i.e. $x = 132/200 = 0.66$
 1 oscillation is completed in 0.66
 Since $T = 1/f$; $F = 1/T = 1/0.66 = 1.52 \text{ Hz}$.
19. A concave mirror of focal length 20cm forms an image Vz the size of the object. Determine the object distance,
- 20cm
 - 60cm
 - 100cm
 - 80cm.
- SOL: Image size is $\frac{1}{2}$ of the object = $11/2$
 $1/f = 1/v + 1/u$
 $1/20 = 1/(U/2) + 1/u$
 $= 1/20 = 2/u + 1/u$, $1/20 = 3/u$
 $U = 60$
20. A pool of Water is 1 2m deep, what is the apparent depth? (Refractive index of water-4/3)
- 1 6.0m
 - 48.0m
 - 9.0m
 - 36.0m
- SOL: Refractive index -real depth = $4 = 12$ apparent dept 3 Ad
 Cross multiply, $36 = 4 \times \text{Ad}$
 $\text{Ad} = 36.0/4 = 9.0$.
21. In the optical systems of the eye and lens camera, which statement is not true
- The ciliary muscle controls the focal length of the eye lens
 - The focal length of the cameral lens is fixed
 - The retina and camera film have similar purposes
 - The focal length of the eye lens is fixed.
- (The iris of the minimi eye lens lie functions is the diaphragm of the 0111101 ; i, while there in a of the human eye has the same functions as the film of the camera).
22. A cell needed 3.0V of the wire of a potentiometer to balance its e.m.f, but only 2.0V to balance its p.d when a resistance of 43Q was connected across the terminals. Calculate the internal resistance of the cell,
- 1.0
 - 0.5
 - 1.5
 - 2.0
- SOL: Using $V_1/V_2 = L_1/L_2 = E_1/E_2$
 $E_1 = V_1$, $0.3 + E = V + 2$
 $0.3 + 1R + 1r = 1r + 2$
 $1r = 2 - 0.3$, $1r = 1.7$
 $r = 1.7/0.5 = 2.4\Omega$
 Recall

$$V = IR$$

$$I = V/R$$

$$I = 2/4 = 0.5$$

23. When a steady current flows in a..... Question not clear

24. A transformer has a primary coil of 400 turns and secondary coil of 200 turns.

If the primary coil is connected to 240V

ac mains, calculate the efficiency given that the current in the primary is 3A and secondary is 5A.

(a) 60% (b) 50% (c) 83% (d) 62.5

SOL: Recall; $1S/1P = EP/ES = NP/NS$

$$200/400 = ES/240, ES = 120V$$

Efficiency = $\frac{\text{output power}}{\text{input power}} \times 100$

$$= \frac{120 \times 5}{240 \times 3} \times 100 = 83\%$$

25. When a p-n junction diode of a semiconductor, is forward biased,

(a) a large current is obtained,

(b) a small current is obtained,

(c) no current is obtained.

(d) the current obtained remains the same.

(In a forward biased semi-conductor, a large current is obtained)

26. Which is the largest single cell in the body?

(a) the neuron (b) the ovum

(c) muscle (d) cell

(The largest single cell in the body is the ovum, while the liver is the second largest organ after the spleen)

27. Blue-green algae belongs to the phylum

(a) cyanophyta (b) schizophyta (c) chlorophyta (d) chrysophyta.

(When green algae and bacteria belong to the kingdom Monera and the phylum Schizophyta).

28. The concept of antibiotics started with the concept of

(a) Gregor Mendel (b) Mary Slessor (c) Louis Pasteur (d) Alexander Fleming.

(Alexander Fleming discovered the antibiotic action of penicillium (fungi) when grown in a cultured bacteria zone. This discovery was somewhat accidental)

29. The botanical name for yellow yam is

(a) Discocarya eschscholtzii (b) Discocarya rotundata (c) Discocarya alata (d) Discocarya dometorum.

SOL: Discocarya eschscholtzii = 1- Yellow yam

Discocarya rotundata = White yam

Discocarya alata = Water yam

30. Which of the following tissues is made up of dead cells?

(a) Xylem vessels (b) Cambium (c) Mesophyll (d) Palisade

(Xylem vessels, mesophyll and palisade are young living tissues in plants. Cambium tissues are dead, thus forming secondary thickening in the stems of dicot plants).

31. Which is not a function of the liver?

(a) Storage of iron. (b) Formation of bile

(c) Breakdown of excess amino acids (d) Excretion of urea from the blood.

SOL: **Sonic basic functions of the liver.**

• Deamination of amino acids to urea and keto acids.

• Storage of iron from the breakdown of haemoglobin i.e. Ferritin

Storage of bile from bilirubin for emulsification of fats.

- Storage of glucose as glycogen, while the kidney excretes urea. NB: Glucose is stored in plants as starch.
- 32. Alkaline pyrogallol was used in an experiment. Thai experiment must have been connected with
(a) respiration (b) photosynthesis (c) fratisphation (d) excretion.
(Alkaline pyrogallol is used to absorb oxygen gas (O_2) in a system, making the non-oxygenated system a control and the oxygenated system a variable. Thus of all the options, only respiration requires oxygen. Nil: Muring respiration, oxygen is utilized, vfiilc during photosynthesis oxygen is evolved as a by-product).
- 33. The enzyme that breaks down sugarcane is
(a) lipase (b) ptyalin (c) invertase (d) peptidase.
(Lipase breaks down fats and oils to fatty acids, Ptyalin is found in saliva and converts starch to maltose. Invertase breaks down disaccharides (omojiosaccharide, e.g. maltose, lactose, sucrose etc.) NB; the sugar found in sugar cane is, sucrose, which is a disaccharide of fructose and glucose, the simple monosaccharide invert sugars. The enzyme for such breakdown is called invertase.
- 34. The largest single bone is the
(a) scapula (b) humerus (c) femur (d) skull
(The largest bone in the body is the thighbone (femur), which articulates with the pelvic girdle at the acetrtbnluin. While the smallest bones in the body are found in the middle ear and they are railed the ear ossicles, which conduct sounds to the oval window).
- 35. Auxins are produced at the
(a) root and stem apices (b) young leaves and nodes (c) flower buds (d) leave apices.
(Auxin is a growth hormone found in the apical region of the meristematic tissues of roots and shoots).
NB: The concentration of Auxins for shoot growth is inversely proportional to mat for root growth.
- 36. The hormone which tones up the muscle of a person in times 01 danger is from the
(a) thyroid gland (b) pancreas (c) adrenal-gland (d) sebaceous gland.
(Thyroid gland secretes thyroxin for growth Ft contains iodine and its deficiency leads to cretinism, dwarfism, myxoieilema and hyperplastic goitre. Pancreas has both hormonal and digestive functions. If secretes insulin and glucagon as hormones for normal blood sugar level and also secretes amylopsin, trypsin and lipase which foster digestion. Deficiency of insulin causes diabetes mellitiis. Adrenal gland secretes an emergency hormone called adrenaline in times of danger, anger, fright, excitement, etc. While sebaceous gland secretes oil in the skin).
- 37. A die! with a high concentration of iodine will probably be needed by a patient suffering from a malfunction of the
(a) thyroid gland (b) adrenal gland (c) nervous system (d)circulatory
(See the solution to question 36).
- 38. The sampling method is most often used in ecological study of habitat because; _____ JSj0clear options.
Reason: The mobile organisms are always randomly distributed in the habitat compared to the non-mobile organisms.
- 39. Which of (he following group of factors is completely abiotic?
(a) Soil, water, bacteria, salinity, (b) Salinity, tide, plankton, humidity, light,
(c) Wind, altitude, (d) Conifers, wind, 1>1, rainfall. turbidity,
(Biotic factors include plankton, bacteria, conifers, animals, etc. While abiotic/actors include Tide, P1 salinity, altitude, humidity, light, water, turbidity, etc).
- 40. Nitrogen fixing bacteria and cowpea demonstrate an ecological succession known as
(a) predation (b) parasitism (c) mutualism (d) commensalisms.
{Cowpea needs nitrogen (atmospheric), whirl) it cannot absorb directly except with the influence of nitrogen fixing bacteria that can convert nitrites to nitrates (the form for absorption) and the cowpea in turn houses the bacteria in its root nodules and feeds it also, This association is termed mutualism).
- 41. Plants (hat live in sally water areas are called

- (a) hydrophytes (b) xerophytes (c) halophytes (d) salinophytes.
(Phytes mean plants; Hydrophytes are found in water, xerophytes are found in dry environments e.g. desert, while halophytes are found in salty water environments).
42. A sample of soil was put into a measuring cylinder and water was added to it. After the mixture was shaken, the cylinder was left undisturbed for one hour. This experiment was probably performed to
(a) compare the capillarity of different samples
(b) find out the relative densities of different soil samples
(c) find out the water retaining capacity of the soil
(d) demonstrate the presence of air in the sample.
SOL: The layer separates into distinct zones on standing, after thorough shaking. Thus it is used to determine the relative densities of different particles in the soil, the gravel, being the densest and the humus, the least dense.
43. A farmer X working in a swamp did not eat any food nor drink any water. Which of these diseases can he not contract?
(a) Bilharzia (b) Malaria (c) Cholera (d) Sleeping sickness
SOL: All of the following; bilharzia, malaria and sleeping sickness can be contracted without eating of food nor drinking of water, since they are not water borne or food diseases. But cholera which is water borne, can be contracted through drinking of contaminated water.
B: Bilharzia → Schistosoma Snail
Malaria – Plasmodium malariae -
Female Anopheles mosquito
Cholera – Vibrio cholerae → Contaminated water
Sleeping sickness Trypanosoma
Tsetse fly.
44. Which of the following element may be added to drinking water to lessen dental decay?
(a) Calcium (b) Phosphorus (c) Fluorine (d) Chlorine
SOL: Calcium - Formation of strong bones and teeth and muscle contraction.
Phosphorus - Formation of strong bones and teeth and for blood clotting. Fluorine - Prevention of dental decay.
Chlorine - Used for microbial destruction (antiseptic).
45. A man with normal haemoglobin marries a woman who has sickle cell haemoglobin. They have a child who has sickle cell trait. Which of the following genotype could be associated with the child's haemoglobin?
(a) SS (b) AS (c) AD (d) AA
SOL: Man AA Woman SS
Cross breed
The child's haemoglobin is associated with AS from the F₁ generation.
46. Trichloromethane is a solvent to remove grease from clothing. How after use is the solvent separated from the grease?
(a) By crystallization (b) By chromatography (c) By distillation (d) By filtration
(The solution is now a mixture of organic compounds which can be separated by distillation due to differences in their boiling points)
47. Arrange the following elements of the second period in the order of increasing atomic radius: Lithium, Beryllium, Boron and Carbon,
(a) Carbon > Boron > Beryllium > Lithium (b) Lithium > Beryllium > Boron > Carbon
(c) Lithium > Carbon > Boron > Beryllium (d) Carbon > Lithium > Beryllium > Boron
SOL: According to the periodic law, atoms are arranged in the periodic table in order of their increasing atomic numbers. No. of protons – no. of electrons! Lithium = 3 electron (2,1), Beryllium = 4 electrons (2,2) Boron = 5 electrons (2,3), carbon = 6 (2,2,2), but along the period, atomic radii decrease with an increase in the atomic number due to the screening

- effects of the electrons and attraction by the protons.
48. Argon is used in a gas filled electric lamp because
(a) it is radioactive (b) it has low pressure (c) it is combustible (d) it helps to prevent oxidation.
(For gases to glow, the voltage across the electrodes must be High and the gas maintained at a lower pressure. Thus for every increase in voltage HIM! decrease in (lie pressure of the gas, the glow gets brighter. This is not the reason for using argon, hut because of its inert nature, thus helps to in the prevention of the electric lump filament against oxidation).
49. If 250cm of a saturated solution of potassiumtrioxonitrate (V) at 30°C produces 3.49 of the dry salt calculate the solubility of the salt at 30°C. [K =39, N-14, O- 16J)
(a) 0.112mol/dm³ (b) 0.112mol/dm³ (c) 0.124mol/dm³ (d) 0.1a4mol/dm³
SOL: Solubility - Mass of solutes/Molar mass of solutes K 1 000/Volume of solvent
Mass of solute - 3.4g
Molar mass of solute - KNO₃ [39 + 14 + (16x3)- 10/g
Volume of so Went = 250cm.
Thus, Solubility- 3.4/101 x 1000/250 = 0.13'ImoLfdm'.
50. Calculate the mass of ZnSO₄ produced when excess of ZnCO₃ is added to 50cm³ of 4.0mol/dm³ H₂SO₄. The equation for the reaction is ZnCO₃ + H₂SO₄ → ZnSO₄ + CO₂ + H₂O [Zn-65, S=32, O-16]
(a) 32.2 (b) 37.5 (c) 38.6 (ci) 31.3
SOL: Molarity - No. of moles of solute/ Volume of solution (cm³) x 1000 Molarity - 4.0mol/ dm³ No. of moles = X mol
Volume of solution = 50cm³
Thus, 4.0 - X/50 x 1000, X = 4.0 x 50/ 1000= 0.2moles. Out No. of moles = Reacting mass/Molar mass.
Reacting mass=Y.
Molar Mass = ZnSO₄ [65 + 32 + (16 x 4)] = 161g. Thus 0.2 moles = Y/161, Y = 161 x 0.2 = 32.2g.

ANSWERS

- 1.A 2.B 3.B 4.A 5.D
6.B 7.A 8.A 9.D 10.D
11.D 12.B 13.C 14.n 15.C
16.B 17.C 18.B 19.B 20.C
21.D 22.n 23.C 24.A 25.B
26.B 27.B 28.D 29.A 30.A
31.D 32.A 33.C 34.C 35.A
36.C 37.A 38.C 39.C 40.C
41.C 42.B 43.C 44.C 45.B
46.C 47.A 48.D 49.D 50.A

SECTION A: USE OF ENGLISH

TIME ALLOWED: 1 HOUR 30 MIN

Comprehensive passage 1

Cloning creates serious issues of individuality. The cloned person may experience concerns about his or her distinctive identity, not only because the person will be in genotype (genetic make up) and appearance identical to another human being, but in this case, because he or she may be twin to the person who is the "father" or "mother" if one can still call them that. What would be the psychic burdens of being the "child" or "parent" of your twin? The cloned individual, moreover, will be saddled with a genotype that has already lived, lie or she will not be fully a surprise to the world.

People will always likely compare a clone's performance in life with that of the original. True, a cloned person's nature and circumstances in life will be different*genotype is not exactly destiny. Still, one must also expect parental and other efforts to shape this new life after the original or at least to view the child with original vision always firmly in mind. Why else then would they clone from the star basketball player, mathematician and beauty queen or even dear old dad in the first place?

Since the birth of dolly, there has been a fair amount of doublespeak on this matter of genetic identity. Experts have rushed in to reassure the public that clone would in no way be the same person, or have any confusion about his or her identity, they are pleased to point out that the clone of film star Julia Roberts would not be Julia Roberts. Fair enough. But one is shortchanging the truth by emphasizing the additional importance of the environment, rearing and social setting: genotype obviously matters plenty. That, after all, is the only reason to clone, whether human beings or sheep. The odds that the clones of the basketball star Larry Sini will play basketball are, I submit, infinitely greater than they are for clones of Lockett Willie shoemaker.

QUESTIONS:

1. The author agrees that
 - (a) genetic characteristics alone cannot determine a person's behaviour
 - (b) people who are cloned from others may have greater professional opportunities,
 - (c) cloning human beings is a greater technological challenge than cloning sheep
 - (d) a formal code of ethical rules will be needed once human beings can be cloned.
2. The author use the word "True" to indicate
 - (a) acknowledgement that the passage's opening arguments are tenuous
 - (b) distrust of those who insist on pursuing cloning..research
 - (c) recognition of a potential counter argument -
 - (d) conviction about the accuracy of the facts presented.
3. The question at the end of paragraph 2 ("why else... first place?") chiefly serves to
 - (a) suggest that some issues are not easily resolved
 - (b) argue for the importance of parents in the life of children
 - (c) imply that cloning might displace more familiar means of reproduction
 - (d) suggest the value perceived in a person who might be selected for cloning.
4. In line 1 of paragraph 3 "For" most nearly means
 - (a) pleasing
 - (b) considerable
 - (c) promising
 - (d) ethical.
5. The author mentions two sports stars in order to
 - (a) distinguish between tasting fame and mere celebrity
 - (b) highlight the need for greater understanding of the athlete's genetic data
 - (c) suggest that athlete's special skills have 'a genetic component

(d) argue against the genetic analysis of any sport star is physical abilities.

In questions 6 and 7, an idiom is underlined in. each of (he sentences, find under each sentences the group of words that give the nearest meaning to the idiom.

6. Let us now get down to brass tacks,
(a) dodge the facts (b) tell a lie p (c) change the iron tacks to brass pie (d) face the actual fact.
(To "get down to brass tacks" means to start to consider the basic 'facts or practical details of something).
7. He talked my head oil when I visited him
(a) wearied me with talking (b) pushed of T his heart
(c) had nothing to say (d) gave himself a headache.

In questions 8 and 9 choose the word or set of words fit best fits the meaning of the sentence as a whole

8. His magazine was ... by the government for an offensive publication
(a) suspended (b) proscribed (c) condemned (d) sealed.
9. The mobile phone ... many times in the last thirty minutes
(a) rang (b) rung (c) been ringing (d) being rung.

In questions 10 and 11, select the option that is nearest in meaning to the underlined word.

10. Can you mention any two immoral acts?
(a) economical (b) dangerous (c) encouraging (d) fantastic..
(“Inimical” means harmful, dangerous etc).
11. Do your parents always decry insubordination?
(a) speak against (b) tolerate (c) embrace (d) overlook.
(The word “decry” means to strongly criticize somebody or something especially publicly)

In questions 12 and 13 choose from the questions given in A - I) the one which is opposite in meaning to the word underlined in each of the following sentences).

12. I know you will prefer a colossal amount as a monthly salaiytoa ... amount even though you are a bachelor,
(a) fantastic (b) meager (c) few (d) huge.
(“Colossal” has to do with “huge”, “large” etc. Its antonym from the above is “meagre”).
13. I think the manager should have embarked on actions that would soothe the feelings of the striking workers instead of uttering statements that could ... them,
(a) incense (b) placate (c) stimulate (d) sensitive.

After each of the following sentences in questions 14 and 15 is a list of possible interpretations. Choose the interpretation most appropriate for each sentence.

14. For all the cared, the match was as good as lost. This means that
(a) he didn't care if the match was lost
(b) he was almost certain that the match would be lost

- (c) he was afraid that the match would be
 (d) he cared very-much about losing the match
15. It is possible that he has not read that book yet. This means that he
 (a) has certainly not read that book yet
 (b) has probably read the book
 (c) may not have read that book yet
 (d) has still not read (hat book.

ANSWERS

- 1.A 2.C 3.B 4.B 5.C
 6.D 7.A 8.B 9.B 10.B
 11.A 12.B 13.C 14.B 15.C

SECTION B: GENERAL PAPER (PHYSICS, CHEMISTRY AND BIOLOGY)

- Which of the following consist of a pair of fundamental quantity only?
 (a) Luminous Intensity, Time (b) Current, Weight (c) Mass, Velocity (d) Length, Speed
 (Fundamental quantities includes Mass, Length, Time, Current, Luminous Intensity, Amount of substance, Temperature etc).
- Which of the following type of motion is least exhibited by the toy top?
 (a) Random (b) Rotational (c) Oscillatory (d) Translational.
- A cube if side 20cm floats in water with 15cm of its height submerged. What is the density of the material of the cube?
 (a) 1.85gincin° (b) 0.85gmcm³ (c) 0.75gmcm³ (d) 0.65gmcm³.
- Which one of the following is not a contact force?
 (a) Gravity (b) Viscosity (c) Tension (d) Reaction.
 (Gravity is a field force, whereas Viscosity, Tension and Reaction are contact forces).
- The motion of the prongs of a turning fork can rightly be described as
 (a) oscillatory (b) translational (c) random (d) rotational.
 (Turning fork is a percussion instrument which exhibits oscillatory or vibratory motion when set into vibration)
- The path described by a projectile motion is
 (a) convex (b) Concave (c) parabolic (d) a straight line.
 (This path is culled trajectory or parabolic as it is a curved motion with acceleration).
- Which of the following sources of energy is renewable?
 (a) petroleum (b) coal (c) nuclear (d) ocean waves.
- Which of the following instruments has a pure
 (a) Vibrating string (b) Turning fork (c) Siren (d) Violin.
- Hot water is added to four times its mass of water at 20°C and the resulting temperature is 30°C. What is the initial temperature of the hot water? (Assume constant specific heat capacity)
 (a) 50°C (b) 60°C (c) 70°C (d) 80°C

SOL; Let the Mass of hot water be = I.

Let the Mass of water be $M_H = 4 \times 1 = 4$. Assumed S.H.C of water - 4200..

Initial Temperature of water = 20°C, Final temperature of water = 30°C.

Using the formula from Newton's law of cooling;

$$\theta_H = M_W C_H O_W / M H C H = 4 \times 4200 \times (30 - 20) / 1 \times 4200 - 40^\circ \text{C}$$

$\therefore$ Initial temperature of Mot water - $40^\circ \text{C} + 20^\circ \text{C} - 60^\circ \text{C}$

Final temperature of hot water = $40^\circ \text{C} + 30^\circ \text{C} - 70^\circ \text{C}$.

10. A certain radioactive source emits radiations that were found to be deflected by. both magnetic and electric fields. The radiations are
(a) infrared rays (b) x-rays (c) gamma rays (d) beta rays,
(There are three types of radioactive emission, which are alpha rays, beta rays and gamma rays. Only alpha and beta rays are deflected by both magnetic and electrical fields, while gamma rays are undeflected).
11. Which does not decolorizes bromine water?
(a) C_2H_2 (b) H_2 (c) C_2H_6 (d) C_3H_6 .
(Decolorization of bromine water is one of the chemical properties of all unsaturated hydrocarbons. Thus all alkenes and alkynes do this except alkanes).
12. What is formed when starch is completely hydrolyzed?
(a) Fructose (b) Glucose (c) Lactose (d) Maltose.
13. Aluminum materials should not be exposed to alkalis because aluminum is
(a) basic (b) acidic (c) amphoteric (d) a reducing agent.
(Amphoteric substances react with acids and alkalis to form salt and water).
14. Sulphur
(a) forms two alkaline oxides (b) is spontaneously flammable
(c) burns with a blue flame (d) conducts electricity in the molten state,
(When Sulphur is heated in plentiful supply of air, it burns with a bright blue flame to form SO_2 with a little amount of SO_3).
15. The bleaching action of chlorine in water is because
(a) of its reducing property (b) of its oxidizing power. (c) it is an oxygen acceptor
(d) it is a weak acid.
16. What change will occur during the electrolysis of brine, using carbon electrodes?
(a) Chlorine will be given off at the cathode
(b) Oxygen will be given off at the anode
(c) Sodium will be deposited at the cathode
(d) The resulting solution after electrolysis will be acidic.
17. A certain mass of gas has a volume of 241cm^3 at 18°C and 753mmHg pressure. What would its volume be at S.T.P?
(a) 224cm^3 (b) 2.24cm^3 (c) 22.4cm^3 (d) 0.0224cm^3
SOL: $P_1 = 753\text{mmHg}$ $V_1 = 241\text{cm}^3$, $T_1 = 18^\circ \text{C} = 18 + 273\text{K} = 291\text{K}$,
 $P_2 = 760\text{mmHg}$ (S.P), $V_2 = ?$
 $T_2 = 273\text{K}$ (S.T),
Using the general gas equation,
 $P_1 V_1 / T_1 = P_2 V_2 / T_2$
 $\therefore V_2 = P_1 V_1 T_2 / T_1 P_2 = 753 \times 241 \times 273 / 291 \times 760 = 224.0\text{cm}^3$.
18. If the solubility of a salt at 25°C is 45g per 100g of water, how much of the salt will be obtained from 50g of the solution?
(a) 20g (b) 22g (c) 25g (d) 2.15g
19. What is not found in an atom?
(a) Electron (b) Proton (c) Molecule (d) Neutron
(An atom is made up of protons, neutrons and electrons).

20. If the P of a solution of dilute hydrochloric acid is given as 2, the concentration of this solution in mol dm^{-3} is
 (a) 0.01 (b) 0.02 (c) 1.0 (d) 2.0
 SOL: $P = -\log_{10}[H^+]$, $\therefore$ if $-\log_{10}P = 2$
 $10^2 = \text{antilog}^{-1} = 0.01 \text{ mol dm}^{-3}$.
21. Hydra is able to perform all the following except
 (a) feeding (b) photosynthesis (c) moving (d) egestion.
 (Hydra performs all the above processes except photosynthesis, which is a major characteristic of chlorophyllous plants).
22. Spherical bacterial cells arranged in chains are called
 (a) staphylococci (b) streptococci (c) cocci (d) bacilli.
 (Spherical shaped bacterial cells are called cocci. When in cluster, they are called staphylococci, when in chains, they are called streptococci and when in group of two, they are called diplococci). NO: Rod like bacterial cells are called bacilli, spiral like ones are called spirillae, while comma shaped ones are called vibrios.
23. One of the following factors that must be considered for safe blood transfusion is
 (a) Social class of the donor
 (b) rhesus factor of the donor and the recipient
 (c) age of the recipient
 (d) nationality of the donor.
24. The theory of use and disuse of organs was propounded by
 (a) Charles Darwin
 (b) Jean Lamarck
 (c) Louis Pasteur
 (d) Robert Hooke.
25. Which of these is not a function of the kidney?
 (a) Production of urine (b) Production of bile (c) Osmoregulation (d) Removal of urea.
 (Production of bile is a function of the gall bladder in the liver).
26. Auxins are produced in the
 (a) petioles of leaves (b) Epidermis of roots and shoots
 (c) Parenchyma of roots and shoots (d) Apical regions of roots and shoots.
 (They are produced at the apical regions (young buds) of roots and shoots). Nil:
 The amount of auxins required for shoot growth is higher than that required for root growth.
27. Which of the following mammalian features acts as a shock absorber to the developing embryo?
 (a) Umbilical cord (b) Amniotic fluid
 (c) Placenta (d) Amnion.
28. The maintenance of a constant internal environment is called
 (a) Environmental regulation (b) Balance in nature (c) Peristalsis (d) Homeostasis.
29. People in the blood group which receive blood from another blood group fall into (a) AB blood group (b) O blood group (c) A blood group (d) B blood group – Answer is B because they are Universal recipients because they have no antigens on their red blood cell membrane, while AB blood group are universal donors in that they have antigens A and B on their membrane.
30. Which of the following crop disease is caused by a fungus? (a) Groundnut Rosette (b) Black pod disease of cocoa (c) Yam Mosaic (d) Cassava blight

ANSWERS

- 1.A 2.A 3.C 4.A 5.A
 6.C 7.D 8.C 9.B 10.D

11.C 12.B 13.C 14.C 15.B
16.C 17.A 18.D 19.C 20.A
21.B 22.A 23.B 24.B 25.B
26.D 27.B 28.D 29. B 30. B

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12. I took time by the forelock and succeeded in getting a scholarship
A) used force B) did a clever thing C) acted promptly D) used a cunning means
13. I laughed up my sleeve as he told the tales of his exploits.
A) was openly making fun of him B) was very pleased C) laughed D) was secretly amused.

INSTRUCTION: From the words lettered A-D, choose the word or group of words that is nearest in meaning to the underlined expression as it is used in the sentence.

14. The rioters despoiled the main markets
A) looted B) destroyed C) burnt D) desecrated
15. What happened last night was a novelty
A) a happy occurrence B) a usual occurrence C) an unusual occurrence
D) a sad occurrence
16. Some people believe that it is ominous for the dog to bark at night
A) giving an evil B) frightful C) sad D) evil
17. In a section of the town many people were living in abject poverty
A) degraded B) despicable C) shameless D) objectionable
18. The contribution of the engineer was priceless
A) invaluable B) worthless C) nice D) exorbitant
19. The young man surmounted all the oppressive measures to break his will
A) overcame B) fought C) renounced D) defeated

Read the passage below and answer questions 20-24 that follow.

The premise behind the question seems to be that if ~~one~~ does not unhesitatingly assert belief in evolution, then one must necessarily believe that God created the world and everything in it in six 24-hour days. But limiting this question to a stark choice between evolution and creationism does a disservice to the complexity of the interaction between science, faith and reason.

The heart of the issue is that we can not drive a wedge between faith and reason. I believe wholeheartedly that there can not be any contradiction between the two. The scientific method, base on reason seeks to discover truths about the nature of the created order and how it operates, whereas faith deal with spiritual truths. The truths of science and faith are complementary: they deal with very different questions, but they do not contradict each other because the spiritual order and the material order were created by the same God.

People of faith should be rational, using the gift of reason that God has given us. At the same time, reason itself cannot answer every question. Faith seeks to purify reason so that we might be able to see more clearly, not less faith supplement the scientific method by providing and understanding of values, meaning and purpose.

More than that, faith-not-science can help us understand the breadth of human suffering or the depth of human love. Faith and science should go together, not be driven apart.

The question of evolution goes to the heart of this issue. If belief in evolution means simply assenting to microevolution, small changes over time within a species, I am happy to say, as I have in the past, that I

believe it to be true. If on the other hand, it means assenting to an exclusively materialistic, deterministic vision of the world that holds no place for a guiding intelligence, then I reject it.

20. In this passage, the writer seeks to
A) portray the superiority of science over faith B) show the compatibility of faith and science
C) show the complexity of evolution and creationism D) show the contradictions of faith and reason
21. The writer's attitude to the subject of the passage could be described as
A) cynical B) ambivalent C) disapproving D) conciliatory
22. From the passage, which of the following does the writer support?
A) evolutionary changes within a small group of organisms B) the creation of the world in a six 24-hour day C) without the scientific method, faith cannot be practice D) those who believe in creation cannot be good scientists
23. A possible title for this passage is
A) the absurdity of faith without reason B) science, faith and reason C) the complementarity of evolution and creation D) a deterministic vision of the world
24. Which of the following idea is NOT true according the passage?
A) both science and faith express various truths B) to believe in evolution is to deny the existence of God C) the world could be better understood by bridging the gap between reason and faith D) both science and faith express various truths

INSTRUCTION: From the words lettered A-D, choose the word or group of words that is nearest in meaning to the underlined expression as it is used in the sentence.

25. Do you know that the _____ is different from the latter?
A) former B) beginning C) latest D) later
26. While some animals are still extant, some are _____ today
A) distant B) infant C) extinct D) domestic
27. Your brother always speaks euphemistic words to address us but it is characteristic of your sister to speak _____ words to us
A) few B) rash C) harsh D) polite
28. While some activities are enervating, some other activities are _____
A) invigorating B) exacerbating C) immunizing D) embarrassing
29. We expect the boy to show remorse for the offence he committed, but he remained _____ to the surprise of all
A) indifferent B) respectful C) unrepentant D) reserved
30. Whereas Bello's behaviour is erratic, Tito's manner of doing things is _____
A) predictable B) allergic C) assertive D) eccentric

GENERAL PAPER

INSTRUCTION: From the words lettered A-D choose the appropriate answer.

Which of the following is not a general characteristic of waves?

- A) diffraction B) polarization C) reflection D) refraction

The thermometric substance of an absolute thermometer is

- A) platinum B) mercury C) alcohol D) helium

Which one of the following has the highest specific heat capacity?

- A) water B) liquid hydrogen C) copper D) lead

Which of the following sources of energy is non-renewable?

- A) tide B) solar C) ocean waves D) biomass

What type of electromagnetic radiation can be used to take photographs in the haze?

- A) radio waves B) visible rays C) ultraviolet rays D) infrared rays

The difference in mass number (A) of atoms of the same element is due to a difference in

- A) proton number B) number of electrons C) neutron number D) proton and electron number

A man hears his echo from a nearby hill 2 seconds after he shouted. If the frequency of his voice is 260 Hz and the wavelength is 1.20 m, how far away is the hill?

- A) 312 m B) 315 C) 335 m D) 340 m

The colour of the positive column in a gas discharge tube is characteristic of the

- A) Gas and the coating in the tube B) pressure of the gas only C) Temperature of the gas only D) pressure and potential difference

Which of the following pairs of particles each have the same number of electrons as argon?

[Ar = 18]

- A) Li^+ and F^- B) Na^+ and F^- C) Na^+ and I^- D) K^+ and Cl^-

A mixture of 0.20 mole of Ar, 0.20 mole of Na and 0.30 mole of He exerts a total pressure of 2.1 atmosphere. What is the partial pressure of He in the mixture.

- A) 0.70 atmosphere B) 0.80 atmosphere C) 0.90 atmosphere D) 0.65 atmosphere

The pH of a solution is 12.23; calculate the hydrogen ion concentration of the solution.

- A) 1.52×10^{-2} B) 1.69×10^{-2} C) 1.74×10^{-2} D) 1.87×10^{-2}

Which of the following is an artificial element?

- A) plutonium B) lanthanum C) uranium D) lutetium

One common reaction may be represented ionically by the equation, $\text{Ba}^{2+} + \text{SO}_4^{2-} \rightarrow \text{BaSO}_4$ this occurs during

- A) Hydrolysis B) Synthesis C) Thermal association D) double decomposition

White phosphorous is stored under water because it

- A) is corrosive B) is corrosive C) reacts with carbon (iv) oxide D) is hydrolyzed by atmospheric moisture

When aluminium tetraxosulphate(vi) reacts with excess sodium hydroxide, the formula of one of the compounds formed is

- A) AlNaO_2 B) $\text{Al}(\text{OH})_3$ C) $\text{NaAl}(\text{OH})_4$ D) Na_3AlO_3

NIGERIAN POST UNIVERSITIES MATRICULATION EXAMINATIONS
LIFE SCIENCE PHAR/DENTISTRY
MEDICINE

AUGUST, 2008

Time allowed: 1 hour

INSTRUCTIONS

Read the following instructions carefully:

1. Use HB pencil to shade your answer. Ensure that any shading in error is thoroughly erased
2. Candidates should write their Names, JAMB Registration numbers and the Question Paper Option given to them in the appropriate space in the Answer Sheet.
3. Write your JAMB Registration numbers on the Question Paper in the space provided at the top of Page 1.
4. Attempt all questions.
5. The use of calculators and/or similar electronic devices is NOT allowed

SECTION A: ENGLISH

From the words lettered A-D choose the word which best completes each of the following sentences.

1. The Solution to the problem is happily, _____
A) at hand B) on hand C) within hand D) near hand
2. My grandmother told a story that made me _____
A) laughing B) laugh C) laughed D) to laugh
3. The principal with all the teachers and students _____
A) has arrived B) were arriving C) have arrived D) were to arrive
4. The triplets love _____ very much
A) each other B) one another C) the other D) the others
5. The man asked if she _____ her face
A) have run B) had ran C) have ran D) had ran
6. Please _____ the light before you sleep.
A) extinguish B) put out C) put off D) switch off
7. The woman _____ the sharp knife from the baby in a subtle way
A) wrested B) rested C) wrestled D) arrested

INSTRUCTION: After each of the sentences below a list of possible interpretations of all or part of the sentence is given. Choose the interpretation that you consider most appropriate for each sentence.

8. The elders felt it was a forlorn hope and vehemently opposed it.
A) a dangerous B) a plan which would not favour the elder C) a planless course of action
D) a plan which had very little likelihood of success
9. The sword of Damocles will be hanging over the council if that party gain control
A) the sword of justice B) imminent danger in the midst of prosperity
C) the man's influence D) the curse of God
10. Bisi stole a march on us by beating the gun at the race
A) gained an advantage B) came last C) came first D) was disqualified
11. Because the Principal still refused his permission his staff finally took the bit between their teeth
A) carried placards B) got out of control C) tore up the timetable D) refused to go to work

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17. An alloy of copper and tin used in electrical contacts is
A) Brass B) solder C) bronze D) type metal
18. Kerosene and petrol (gasoline) are
A) hydrocarbons with kerosene having bigger molecules B) hydrocarbons with petrol having bigger molecules C) hydrocarbons with the same size molecules but of different shape D) compounds of carbon and hydrogen, and sulphur for kerosene, but nitrogen for petrol
19. The preparation of ethoxyethane ($C_2H_5OC_2H_5$) from ethanol can be considered as
A) dehydration B) Oxidation reactions C) hydrolysis reactions D) dehydrogenation reactions
20. Which of the following gives the correct formula of the fatty acid present in palm oil?
A) $CH_3(CH_2)_{14}COOH$ B) $CH_3(CH_2)_{16}COOH$ C) $CH_3(CH_2)_{15}COOH$ D) $CH_3(CH_2)_{17}COOH$
21. Glycerol had the typical properties of
A) an acid B) a base C) an alcohol D) an ester
22. A certain natural product contains the ester glycerol palmitate soap can be made from this by
A) passing over activated charcoal B) Boiling with sodium hydroxide C) mixing with hydrogen D) heating with glycerine
23. Which of the following is NOT a thermosetting plastic?
A) terylene products B) phenol - methanol products C) carbamide - methanol products D) melamine - methanol products
24. The trajectory of a projectile motion is
A) ellipsoidal B) parabolic C) parabolic D) spherical
25. A body accelerates from rest with an acceleration of 8ms^{-2} for 10 seconds. Calculate the distance covered in the 10th second
A) 76m B) 97m C) 324m D) 400m
26. Which of the following infection is unlikely to respond to antibiotics treatment?
A) tetanus B) measles C) typhoid fever D) ringworm
27. Pest can be controlled without the subsequent danger of polluting the ecosystem by
A) spraying the ecosystem with pesticide B) keeping the population of natural enemies constant C) increasing the population of natural enemies of the pest D) eliminating the natural enemies of the pests
28. Autotrophs are also describes as
A) consumers B) carnivores C) producer D) decomposers
29. Which of the following bones is Not a component of the appendicular skeleton?
A) tibia B) sacrum C) radius D) humerus
30. If two parents are sickle cell carriers then their genotypes would be
A) $Hb^A Hb^A$ and $Hb^A Hb^A$ B) $Hb^S Hb^S$ and $Hb^S Hb^S$ C) $Hb^A Hb^A$ and $Hb^S Hb^S$ D) $Hb^A Hb^S$ and $Hb^A Hb^S$
31. Prokaryotic cells are found in
A) monerons B) algae C) tracheophytes D) spirogyra

31. The correct order of classification of human being is
 A) animalia, chordate, vertebrata, mammalie, primate, hominidate, homo and sapiens
 B) animalia, vertebrata, mammalie, chordate, primato, homo, sapiens and hominidee
 C) chordate, animalia, vertebrata, mammalie, primate, hominidate, and sapiens
 D) Animalia, chordate, vertebrate, primate, mammalian, homo and sapiens
32. Deficiency of sulphur causes
 A) poor, weedy growth become slender
 B) extremely stunted growth
 C) yellowing of leaves stems
 D) weak, stunted growth
33. Vitamin K is important in the
 A) formation of enzymes involved in synthesis of amino acid
 B) formation of red blood cells
 C) synthesis of blood clotting factors
 D) formation of co-enzymes involved in cellular respiration
34. The following are the components of nucleotides except
 A) a five-carbon sugar
 B) a six-carbon sugar
 C) a phosphate group
 D) an organic nitrogen-containing
35. Which of the following inherited conditions usually shows incomplete dominance?
 A) baldness
 B) tongue rolling ability
 C) haemophilia
 D) sickle cell anaemia
36. which of the following is not a theory under Darwin's theory of organic evolution
 A) over production
 B) struggle for existence
 C) mutation
 D) offspring show variations
37. Which of the following does not cause mutation?
 A) x-rays
 B) mutant
 C) ultra-violet light
 D) cosmic radiation
38. Which of the following organisms has homodont dentition?
 A) a lizard
 B) sheep
 C) cat
 D) man
39. Which of the following habitats cannot be used for the study of succession?
 A) a pond
 B) savanna grassland
 C) well cultivated land
 D) abandoned farm land
40. Which of the following statement is/are correct
 i. one mole of sodium atoms contains 6.02×10^{23} atoms and weighs 23g
 ii. one mole of magnesium atoms contains 6.02×10^{23} atoms and weighs 24g
 iii. one mole of oxygen atoms contains 6.02×10^{23} atoms and weighs 16g
 A) i and ii
 B) ii and iii
 C) i and iii
 D) i, ii and iii

2. A
3. B
4. A
5. B
6. D
7. B
8. A
9. B
10. B
11. A
12. B
13. C
14. D
15. A
16. C
17. D
18. B
19. A
20. A
21. B
22. D
23. A
24. C
25. B
26. A
27. C
28. C
29. A
30. C
A

1. B
2. D
3. B
4. D
5. D
6. C
7. A
8. A
9. D
10. C
11. B
12. A
13. D
14. B
15. C
16. C
17. A
18. A
19. A
20. C
21. B
22. A
23. C
24. A
25. B
26. C
27. C
28. B
29. D
30. A
31. A
32. C
33. C
34. B
35. A

36.
37.
38.
39.
40.

C
B
A
C
D

JAMB Registration Number

OPTION D

UNIVERSITY OF BENIN, BENIN CITY, NIGERIA
POST UNIVERSITIES MATRICULATION EXAMINATION SCREENING
WEDNESDAY, AUGUST 26th 2009, 9AM

Time allowed: 1 hour

INSTRUCTIONS

Read the following instructions carefully:

- Use **HB pencil** to shade your answers. Ensure that any shading in error is thoroughly erased.
- Candidates should write the **Question Paper Option** given to them in the appropriate space in the **Answer Sheet**.
- Write your **JAMB Registration numbers** on the **Question Paper** in the space provided at the top of Page 1.
- Attempt all questions.
- The use of calculators and/or similar electronic devices is **NOT** allowed.

From the options A – D, choose the word that has the same sound as the one underlined.

1. Poor.
A. More B. Tore C. Dour D. Door
2. Great
A. Cake B. Break C. Team D. Beam
3. Cloth
A. Cloth B. Tooth C. Wordy D. Forth
4. Sing
A. Goose B. Game C. Banged D. Jungle
5. Machine
A. Chancellor B. Bishop C. Character D. Measure

From the options A – D provided, choose the one nearest in meaning to the underlined word(s).

6. Many tourists say this country has a bracing climate.
A. harsh B. invigorating C. benevolent D. same
7. After the race, some of the athlete felt completely done in.
A. strained B. satisfied C. disappointed D. exhausted
8. My sister's husband is very fastidious about his food.
A. careful B. hygienic C. ravenous D. particular
9. Because the student failed to obey the regulation restricting movement in the school compound, he was expelled.
A. answer B. observe C. notice D. understand
10. Peter's breathing soon became difficult in the thin air of the hills, and there had to be regular pauses for regulation.
A. rewarded B. rest C. recovery D. restoration

From the words lettered A-D choose the interpretation that gives the correct meaning of the idiom in the sentence or phrase.

11. The teacher said that it was bad to cook the books. This means to...
A. write good books for the public to read and enjoy B. falsify figures dishonestly C. willfully destroy a document which contains vital information D. publish books as a means of living.

He is in the first flight of footballers. This means he is...
A. in the first team B. always the first footballer to score a goal C. among the best footballers D. the first footballer to run away from camp.

13. To say that 'a rolling stone gather no moss' means, A. an unsteady person is bound to encounter failure B. A young person cannot succeed easily in life C. A person who runs every time cannot live long D. It is not good to make a stone roll.

Osazee scored a brilliant hat trick in the final match of the world cup fiesta. This means that he...
A. Won the price for the best player B. Failed to score the winning goal C. Had a hat on during the game D. Scored three successive goals in the match

15. The judge 'had his hands tied' in the case in which his son was arraigned for a charge of murder. This means that the judge...
A. Was withdrawn from the case B. Was handcuffed while the trial lasted C. Could not help convicting his son D. Was conspicuously partial.

In the following sentences, choose from the options A – D that best completes the sentences

16. Among the books in the library, he found Bible.
A. an old family brown leather B. an old brown leather family C. a brown leather old family D. an old leather brown family.
17. The poor man lived in house.
A. an old mud-brick large broken-down B. a large old mud-brick broken-down C. a large old broken-down mud-brick D. a broken-down old large mud-brick.

Life

18. The man donated a ... machine to our school. A. Modern duplicating large red B. Modern large red duplicating C. Large modern red duplicating D. Red duplicating large modern.
19. The beauty pageant was won by a ... girl. A. beautiful dark and tall B. tall dark and beautiful C. dark tall and beautiful D. beautiful tall and dark.
20. The house ... the kidnappers were kept is being watched by the police. A. which B. where C. in where D. of which
21. A respectable person should not ... on an agreement. A. renege B. disagree C. honour D. dispute
22. The little boy ... to the occasion when he was asked to make a speech at his mother's funeral. A. rose B. lifted C. stood D. fell
23. Do you know the story about the ... A. the president's of Nigeria plane B. the president's plane of Nigeria C. the president of Nigeria's plane D. the plane of the president's of Nigeria.

Read this passage carefully and then answer the questions that follow.

There is always a hustle and bustle of people in a typical city, especially at closing period when shops and offices were closing and workers were hurrying home, some in their private cars, others trying to catch a commercial vehicle, while those who couldn't stand the rush to enter a bus or a cab jump on motorcycles popularly called Okada. It was at the peak of that rush hour that Oshoke arrived in Lagos.

Oshoke had never seen so many people on the streets or so many vehicles on the road, and so when he dropped at the popular Yaba bus-stop, he felt suddenly lost like a man lifted from the ground without any warning or explanation.

Another observation that stupefied him was not just the number of motor-cycles that move like a swarm of ants, but the way each rider appears to be oblivious of any other vehicle or human being around them, posing a grave danger to life.

In his stupor, somebody tapped his shoulder and all of a sudden, he felt empty, the way a young woman feels when her skirt unexpectedly falls in the middle of a busy street. But he managed to turn to meet the hard cold eyes of a man glaring at him. "What is the problem?" the voice reminded him of a lion's roar. As Oshoke later admitted it was perfect voice with which to wake a dead man.

Oshoke was later to learn from the stranger that the city had the worst drivers even though drivers generally in the cities, see every other driver as a rival on the road - an enemy to be conquered. One minute they would be overtaking their cars, the next stopping in the middle of the road to pick-up fares without any signal to other drivers or road users of their intentions.

pedestrians seemed always in danger of their lives as they had to run across the road whenever there was a lull in the traffic.

Life

24. How did Oshoke feel as he arrived in Lagos? A. he was so elated that he felt he was walking on air B. he became so nervous at once C. he was simply dazed D. he was brimful of composure
25. Why was Oshoke bewildered? A. he thought the stranger would attack him B. he suddenly felt cold C. the rowdiness of Lagos D. he was frightened because of the car and people in Lagos.
26. According to the passage, the commuters A. were confused B. were very wicked C. were irrational D. lack discipline.
27. According to the passage, A. passengers are always in danger of crossing the road B. fares are always in danger of being knocked down C. people are always in danger crossing the road D. people are always crossing the road.
28. From the passage, we can infer that all road users. A. need education B. must be directed C. need the right orientation D. need to be monitored by the police.

From the words lettered A-D choose the word or group of words that is opposite in meaning to the underlined word or group of words.

29. Girls these days dress provocatively but the culture requires ... dressing. A. Prodigious B. Preventive C. Modest D. Productive
30. He did the work perfunctorily instead of ... A. perfectly B. carefully C. permanently D. especially

GENERAL PAPER

INSTRUCTION: From the words lettered A-D choose the appropriate answer.

1. Which of the following best describes viruses?
 - A. they grow only inside living cells
 - B. they possess only characteristics of living things
 - C. they can grow on dead cells
 - D. they cannot be crystallized.
2. Loss of nitrogen in the soil is caused by a process known as
 - A. putrefaction
 - B. nitrogen fixation
 - C. denitrification
 - D. nitrification
3. Which of the following represents the Watson-Crick DNA base pair model?
 - A. G-T
 - B. T-C
 - C. T-A
 - D. G-A
4. The liver performs all the following functions EXCEPT
 - A. regulation of heat
 - B. formation of cholesterol
 - C. removal of red blood cells
 - D. synthesis of vitamins
5. All but ONE of the following are atmospheric pollutants.
 - A. diesel
 - B. carbon II oxide
 - C. sulphur (iv) oxide
 - D. hydrogen sulphide
6. On what part of the tongue do we perceive salty taste?
 - A. front
 - B. sides
 - C. back
 - D. sides and front
7. Maintenance of homeostasis in mammals is a function of all the following EXCEPT
 - A. ovaries
 - B. skin
 - C. lungs
 - D. kidneys
8. unicate Biotechnology has helped to produce plants and animals of these desirable features, EXCEPT
 - A. greater resistance to pests and diseases
 - B. higher susceptibility to infections
 - C. better nutritional value
 - D. early yielding varieties
9. Insects which undergo complete metamorphosis are those that
 - A. have the four distinct stages in their life cycle
 - B. do not have the larval stage
 - C. have larva and pupa which resemble the adult
 - D. their eggs hatch to produce nymphs
10. The type of storage organ found in ginger is called
 - A. bulb
 - B. cutting
 - C. tuber
 - D. rhizome
11. The phototropic response of a seedling is detected with
 - A. a clinostat
 - B. a Thermostat
 - C. a Secchi disc
 - D. an auxanometer
12. The substrate in cellular respiration is
 - A. sucrose
 - B. glucose
 - C. glycogen
 - D. adenosine diphosphate
13. A good example of roughage - containing food material is
 - A. orange juice
 - B. corn flour
 - C. Corn
 - D. egg yolk
14. The World Health Organisation defines good health as
 - A. a state of physical and mental well-being
 - B. the absence of deformities
 - C. absence of disease
 - D. a state of physical, mental and social well-being
15. The stage at which chromosomes move towards to opposite poles during mitosis is
 - A. prophase
 - B. metaphase
 - C. telophase
 - D. anaphase
16. Which of the following can sublime
 - A. water
 - B. ink
 - C. iodine
 - D. hydrogen
17. A certain compound on analysis gave 2.02g of carbon, 0.35g of hydrogen and 2.68g of oxygen. If the molecular mass is 90, calculate the molecular mass
 - A. CH_2O
 - B. $\text{C}_2\text{H}_4\text{O}_2$
 - C. $\text{C}_3\text{H}_6\text{O}_3$
 - D. $\text{C}_4\text{H}_8\text{O}_4$
18. Isotopy is due to differences in the number of
 - A. atomic mass
 - B. molecular mass
 - C. protons
 - D. neutrons
19. Chlorine is a ...
 - A. colourless gas
 - B. greenish liquid
 - C. yellow gas
 - D. greenish yellow gas
20. Calcium carbide reacts with water to form ethyne. What mass of calcium carbide (grams) would react to give 1.12dm^3 of ethyne at STP? [Ca = 40, C = 12, H = 1]
 - A. 60g
 - B. 6.4g
 - C. 3.2g
 - D. 3.2g
21. 5.3g of silver trioxonitrate (V) was dissolved in 50cm^3 of water at 25°C . Calculate the solubility of the salt in mol dm^{-3} [Ag = 108, N = 14, O = 16]
 - A. 0.512 mol dm^{-3}
 - B. 0.031 mol dm^{-3}
 - C. 0.062 mol dm^{-3}
 - D. 0.623 mol dm^{-3}
22. Which of the following decomposition equations are correct?
 - A. $\text{Ag}_2\text{CO}_3 \rightarrow \text{Ag}_2\text{O} + \text{CO}_2$
 - B. $2\text{PbCO}_3 \rightarrow 2\text{PbO} + 2\text{CO}_2 + \text{O}_2$
 - C. $2\text{NaNO}_3 \rightarrow 2\text{NaNO}_2 + \text{O}_2$
 - D. $\text{Al}_2(\text{CO}_3)_3 \rightarrow \text{Al}_2\text{O}_3 + \text{CO}_2$
23. In the reaction of aqueous solution of HCl and NaOH, the heat change is heat of
 - A. solution
 - B. neutralization
 - C. formation
 - D. displacement
24. Which of the following gas is colourless and odourless?
 - A. NO_2
 - B. N_2O
 - C. O_2
 - D. b and c
25. The two hydrocarbon used as lower and upper standards for petrol are
 - A. isooctane and benzene
 - B. heptane and 2,2,6-trimethylpentane
 - C. heptane and 2,2,4-trimethylpentane
 - D. heptane and neopentane
26. Benzene commonly undergoes ----- reactions
 - A. addition
 - B. elimination
 - C. substitution
 - D. amendment

27. Under high pressure, real gases deviate from compliance with gas laws because their molecules
A. experience repulsive forces B. have become more energetic C. have become less energetic D. collide inelastically
28. Which of the 3d, 4s, 4p, 4d has the lowest energy? A. 3d B. 4s C. 4p D. 4d
29. One of the causes of algae growth in rivers is the presence of
A. heavy metals B. pesticides C. Phosphates from detergents D. High particulate loading
30. Phenolphthalein solution at pH 2.5 is
A. yellow B. pink C. red D. colourless
31. Which of the following is a vector quantity? A. Distance B. Speed C. Energy D. Weight
32. Calculate the work done by a bricklayer who lifted cement block of mass 10kg from the floor to a height of 2.5m (acceleration due to gravity = 9.8 ms^{-2}).
A. 294.0 J B. 245.0 J C. 30.0 J D. 25.0 J
33. Find the potential energy of a diver of mass 75kg at a height of 10m above a swimming pool (acceleration due to gravity = 9.8 ms^{-2}).
A. 7.35KJ B. 9.41KJ C. 8.82KJ D. 0.75KJ
34. Determine the wavelength of fundamental note of a guitar string of length 65cm long
A. 65cm B. 260cm C. 195cm D. 130cm
35. Find the combined resistance of two wires each of resistance 8-ohms when they are connected in parallel
A. 16-ohms B. 4-ohms C. 20-ohms D. 12-ohms
36. A force of 300N was used to pull a load of mass 40kg at a steady speed up an inclined of length 6.2m and vertical height 1.5m. Determine the velocity ratio
A. 0.57 B. 1.31 C. 1.37 D. 1.77
37. Which of the following is NOT TRUE about pressure
A. Unit of pressure is Nm^{-2}
B. Pressure in a liquid increases with depth
C. Pressure at any point in a liquid acts in all directions
D. Pressure is the ratio of force acting normally per unit area.
38. 120cm³ of gas was collected at 20°C and 745mm of mercury pressure. Find the volume of gas at standard temperature and pressure
A. 1605.7cm³ B. 917.8cm³ C. 305.1cm³ D. 109.6cm³
39. A metal of length 30cm is heated until its temperature rises by 60°C to a new length of 30.04cm. Calculate its linear expansivity.
A. $4.44 \times 10^{-5} \text{ K}^{-1}$ B. $2.22 \times 10^{-5} \text{ K}^{-1}$ C. $3.33 \times 10^{-5} \text{ K}^{-1}$ D. $1.52 \times 10^{-5} \text{ K}^{-1}$
40. Which of the following statements is NOT correct?
A. Molecules of a liquid are stationary B. Atoms combine to form molecules C. The molecules of matter are in constant motion D. Brownian motion is an evidence of particle nature of matter.
41. Determine the elastic constant of a spring if a force of 1N stretches it by 2.7cm *answer is B*
A. 0.4 Nm^{-1} B. 40.4 Nm^{-1} C. 37.0 Nm^{-1} D. 40.0 Nm^{-1}
42. Which of the following objects is NOT in stable equilibrium?
A. A car with wide base and low centre of gravity B. A ball in the middle of a bowl C. A car with narrow base and high centre of gravity D. A cone resting on its base
43. Which of the following is not true of boiling?
A. Temperature is constant during boiling
B. Occurs throughout the entire volume of the liquid C. Takes place at all temperatures
D. It is change from liquid to vapour at the boiling point.
44. A swimming pool viewed directly from above its surface appears to be 12m deep. Find the real depth. (Refractive index of water = $\frac{4}{3}$).
A. 16m B. 20m C. 10m D. 24m
45. A transformer has 500 turns in the primary coil and 400 turns in the secondary coil. Find the voltage in the secondary coil when the primary coil is connected to a 220V main.
A. 132V B. 160V C. 176V D. 182V

$$\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$65 \times \frac{1}{2}$$

$$\frac{65 \times 1}{2} = 32.5$$

radical notation